

A project report on

FEDADAPT: ADAPTIVE FEDERATED LEARNING WITH RIEMANNIAN MANIFOLD REPRESENTATION AND ATCNET FOR MOTOR IMAGERY EEG CLASSIFICATION

Submitted in partial fulfillment for the award of the degree of

Master of Computer Applications

by

SURYA K (24MCA1032)



VIT[®]

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

April, 2026

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School of Computer Science and Engineering

DECLARATION

I hereby declare that the thesis entitled “FEDADAPT: ADAPTIVE FEDERATED LEARNING WITH RIEMANNIAN MANIFOLD REPRESENTATION AND ATCNET FOR MOTOR IMAGERY EEG CLASSIFICATION” submitted by me, for the award of the degree of Master of Computer Applications, Vellore Institute of Technology, Chennai is a record of Bonafide work carried out by me under the supervision of Dr. Anandan P.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Chennai

Date: 08.04.2026

Signature of the Candidate



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CERTIFICATE

This is to certify that the report entitled **“FedAdapt: Adaptive Federated Learning with Riemannian Manifold Representation and ATCNet for Motor Imagery EEG Classification”** is prepared and submitted by **Surya K (24MCA1032)** to VIT Chennai, in partial fulfillment of the requirement for the award of the degree of **Master of Computer Applications**, programme and as a part of PMCA699J - Dissertation II is a bonafide record carried out under my guidance. The project fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

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ABSTRACT

By decoding neural activity linked to specific cognitive tasks, Brain-Computer Interfaces (BCIs) based on Electroencephalography (EEG), allow the brain to communicate directly with the outside world. One of the most frequently studied BCI paradigms is motor imagery classification due to its applications in assistive technology and neurorehabilitation and human-computer interaction. However, the reduced amount of variability across subjects, and variability over time, with EEG means developing generalizable machine learning models from the data poses a challenge. Additionally, when using centralized training approaches in BCIs, sensitive neural data must often be collected from multiple subjects, raising issues of privacy and data-sharing. To overcome these difficulties, this study proposes FedAdapt, a federated learning framework for motor imagery classification that utilizes Attention-based Temporal Convolutional Networks (ATCNet) with Riemannian manifold-based feature representation. In this proposed approach, the EEG trials are first transformed into covariance matrices to represent the spatial relationship between the EEG channels. Covariance matrices exist in the space of symmetric positive definite matrices; therefore Riemannian manifold techniques are employed to convert covariance matrices to tangent space representations appropriate for machine learning models. The transformed data are inputted into the ATCNet architecture, which uses temporal convolutions and attention mechanisms to effectively learn temporal dependencies found in EEG signals. Everyone provides his/her EEG data in a federated manner as an independent client, thus allowing for collaborative learning without the need to share original EEG recordings with the other working group members. Also, a two-phase training procedure that incorporates personalization is introduced during the learning process; this is accomplished by globally training all shared layers of the network and locally fine-tuning subject-specific layers. To evaluate the performance of the proposed method, an experiment using the BCI Competition IV Dataset 2a was performed, showing that the FedAdapt architecture can be used to create global representations of motor imagery patterns while at the same time allowing for improved accuracy through local personalization for subject-specific data sets. Our findings illustrate how the combination of Riemannian geometry, deep learning architectures, and federated learning can be exploited to develop EEG-based BCI systems that are both scalable and privacy-preserving.

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LIST OF ACRONYMS

BCI	Brain Computer Interface
EEG	Electroencephalography
MI	Motor Imagery
FL	Federated Learning
HFL	Horizontal Federated Learning
RA	Riemannian Alignment
IID	Independent and Identically Distributed
CNN	Convolutional Neural Network
ATCNet	Attention-based Temporal Convolutional Network
SCAFFOLD	Stochastic Controlled Averaging for Federated Learning
FedBS	Federated Balanced Sampling
FedNova	Federated Normalized Averaging
FedAvg	Federated Averaging
FedAdapt	Adaptive Federated Aggregation Strategy

Chapter 1

Introduction

1.1 INTRODUCTION

Brain-Computer Interfaces (BCIs) are systems that allow for direct interaction between the human brain and external devices by turning neural signals into movements. One of the most used types of BCI is motor imagery (MI), in which users can create a pattern of EEG signals by simply imagining performing a specific motor activity (e.g., moving the left or right hand). These EEG signals can then be processed using machine learning techniques and classified to predict intended actions. However, there are challenges in creating reliable EEG based BCIs due to the high variability and noise associated with brain signals. EEG signals recorded from different people will generally follow different statistical distributions, which makes it difficult to create models that can be trained on many people but perform consistently across all subjects. This problem is especially pronounced when using multiple subject datasets such as those provided in the BCI Competition IV Dataset 2a, where each subject will have distinctly different brain dynamics.

To tackle the many challenges presented by the use of EEG as a BCI, advanced concepts from areas such as distributed learning, machine learning and geometry have been researched during the last few years. One of the fundamental ideas used in this work has been the use of Riemannian geometry of symmetric positive definite (SPD) matrices, which is a form of geometry used to analyze data sets constructed with covariance matrices derived from EEG data. By utilizing the Riemannian geometry of these SPD matrices in constructing measurements of similarity and defining features for classification, much more meaningful measurements of similarity and feature representations can be achieved compared to those obtained using the Euclidean approach. Federated Learning (FL) is a new way of creating a machine learning model that is able to be used by many people while keeping their private information secure. Through FL, multiple clients (for example, different people or companies) can use multiple models to create one model without having

to share the original data with each other; in this case, the model update is sent to a central server where the new model is created, and this helps keep the data private while making use of all of the knowledge from the multiple clients.

The use of deep learning models with neural signals such as EEGs has also demonstrated great success in identifying complex relationships between EEG signals; for example, an architecture called the Attention-Based Temporal Convolutional Network (ATCNet) has shown the ability to learn temporal and spatial continuity (or coherence) of EEG signals that are created During Motor Imagery. Unfortunately, when the multiple clients use FL, the training process becomes more difficult due to the non-identical distribution (or non-IID) of the data across all clients that make up the FL environment. Recently, there has been increasing interest in using hybrid learning models; these include the combination of classical machine learning models with quantum computing models. Quantum machine learning uses Quantum-inspired transformations and quantum circuit models to represent Complex Nonlinear Relationships between multiple variables across a higher-dimensional space or dimension. The use of quantum models with classical model will help to create improved representations of the features and improved classification of the features for difficult data processing problems such as EEGs for BCIs.

This research study looks at how to apply Deep Learning Models, federated learning and Riemannian Geometry-Based Feature Representations to identify Motor Imagery from the data of multiple EEGs collected from different People and creates the opportunity to look at Quantum Improvement of These Learning Mechanisms.

1.2 OVERVIEW

Motor imagery classification using EEG signals continues to be a difficult task, because of the nature of the signals, which have a low signal-to-noise ratio, are variable between people, and there is little publicly available labelled data to train the classifiers on. Algorithms in machine learning require the collection of data from many individuals at a central place to create large numbers of data members; thus, there are potential privacy,

ownership and ethical treatment issues with regard to the storage and handling of biomedical data. Federated learning is a collaborative method where parties maintain their own records of EEG data but can work together to train models from that data without sharing their recordings. However, one of the main obstacles for federated learning in BCI applications is that the EEG distributions are very heterogeneous across all subjects. Thus, by having different types of hours of work, there is an increase in amount of error when laying the models on all subject's EEG data, which will diminish the overall performance of the averaged model of the subjects combined EEG data. In addition, many conventional methods for classifying EEG use Euclidean features to represent the EEG data, so they do not fully represent the geometry of the covariance matrices from the data. One mathematical approach to modelling these data in a more robust way is using Riemannian Geometry.

Additionally, recent advances in using hybrid Classical and Quantum Learning models to improve the representation of non-linear features and improve classification results for nonlinear biomedical signal processing problems provide opportunities to make further improvements to the performance of these classification methods. This research is motivated by the above-mentioned opportunities and barriers, and it proposes a unified framework between these different elements, namely using Riemannian manifolds to model the EEG data features; building upon that framework in using deep neural architectures to create an EEG classifier; and further, using federated learning to allow more generalisable and robust classifiers of Subject Motor Imagery Classification across multiple Subjects.

1.3 PROBLEM STATEMENT

Many obstacles remain unsolved in motor imagery classification using EEG. For example: Variability between the EEG signals of different people makes it difficult to produce a model which works well across many people.

- The fact that centralized training methods need access to raw data from people's brains raises issues of privacy and sharing of that data;
- Standard Euclidean vectors, which are used as input features, may not represent the

underlying structure of covariance-based EEG features, thus diminishing the ability to classify properly;

- Non-identical distributions of the training data among federated learning clients can impede global aggregation performance.

Therefore, we need to create an efficient way to process data that adheres to all three aspects of privacy, inter-subject variability, and feature representation while achieving high accuracy in classifying motor imagery in BCI applications.

1.4 RESEARCH OBJECTIVES

This investigation has as its main goal to design an advanced learning framework for motor imagery classification from electroencephalography (EEG) data using a combination of the distributed learning paradigm and spatial correlation encoded through geometry. Below are the detailed specific objectives of this project:

- To assess the use of Riemannian manifold-based feature representations in representing the spatial correlations of EEG data;
- To create and develop a deep learning model to extract both temporal and spatial features of EEG data;
- To construct a federated learning model to enable joint training of multiple users while maintaining the privacy of their respective data;
- To measure the effect of personalized local adaptation techniques to improve performance for each user in the federated learning setting;
- To determine the feasibility of improving nonlinear feature representation and therefore classification performance using quantum-enhanced learning processes.

1.5 SCOPE OF THE STUDY

The aim of this study is to investigate the creation and assess the results of an approach to classify motor imagery using EEG signals, using a Federated Learning framework. BCI Competition IV Dataset 2a is the principal source of data for this research. This dataset includes EEG signals from various subjects performing motor imagery tasks. This study will focus specifically on:

- Processing and predetermined features based on EEG signals
- Using Riemannian geometry to represent EEG features using covariance
- Creating deep learning architectures for classifying motor imagery
- Creating a federated learning framework to enable decentralized training of models across participants or subjects
- Classifying and assessing the models' performance by utilizing standard classification metrics

This study only performs an offline experimental evaluation and only utilizes publicly available datasets; this study does not conduct real-time BCI implementation or clinical trials.

1.6 LIMITATIONS OF THE STUDY

The model we propose aimed to help with the categorization of motor imagery within federated environments, yet there remain a number of limitations that should be noted. These limitations are as follows:

- The data from our experiment was pre-recorded and therefore may not be representative of actual variability from real-time BCI systems.
- Our expectation for the federated learning experiments conducted were that they would be performed within a controlled simulated environment, which fails to account for any delay (due to communication) and hardware mismatch.
- The introduction of components such as quantum-enhanced learning requires some level of reliance on the simulation of the quantum circuit being performed on classical hardware, rather than utilizing a quantum computer.
- The primary data set utilized to train and test the proposed model in this research was a single benchmark data set. In turn, this may negatively affect the generalizability of the outcome results to other types of BCI or EEG data sets.

Despite the above limitations, we feel this study supports the feasibility of using geometric representations; deep learning; and federated learning to improve the classification of motor imagery across multiple EEG-based BCI systems (i.e., distributed systems).

Chapter 2

Background

2.1 LITERATURE SURVEY

The Article "Federated GNNs for EEG-Based Stroke Assessment" by A. Protani et al., outlines a federated framework comprised of Federated Learning and Graph Neural Networks to perform stroke severity assessments utilizing EEG signals from multiple institutions while preserving the privacy of these EEG signals. This study models EEG signals as a regression problem and utilizes the NIH Stroke Scale (NIHSS) as the outcome variable. In addition, by modeling the brain's connectivity via a graph/domain, federated models are able to identify complex spatial relationships between the EEG signals from different patients without sharing EEG signal data.

- Advantage - The mean absolute error (MAE) recorded is 3.23 which is comparable to human expert assessments of stroke severity; additionally, the models have the capacity for post-hoc explainability via EdgeSHAP which explains the models' decision-making processes.
- Disadvantage - The models require high-quality multi-channel EEG data and stable communications protocols (such as MQTT) to communicate between institutions; both of which may be difficult to obtain in multiple low-resource clinical settings.
- Methodology - The authors implement a multi-layer GNN architecture combined with Federated Averaging (FedAvg) which provides the framework for explainability via EdgeSHAP.
- Results - The authors found an MAE of 3.23 using a dataset of 72 stroke patients.
- Dataset - The dataset includes EEG data from multiple Italian hospitals; specifically, the authors used EEG signals collected from four specific hospitals.

The paper titled "Exploring Federated Learning for Emotion Recognition on Brain-Computer Interfaces" [2] by Sofia Mongardi and Pietro Pinoli focuses on using Federated

Learning (FL) techniques to recognize emotions in subjects by predicting the level of valence and arousal across multiple users using a wearable brain-computer interface (BCI) system. This study used a decentralized training approach through the implementation of machine-learning algorithms (Random Forest Regressor) to investigate whether collaborative modeling approaches can circumvent the issues associated with non-homogeneous small data sets in the context of developed, rich data on different subjects whilst maintaining individual users' privacy. By simulating an environment where users could wear a limited number of EEG sensors, the authors performed an evaluation of how the tasks performed by the researcher (through simulation) and the investigators (through the use of existing data) that exist in the area of affective computing might have different inherent trade-offs based on both the three different configurations under which they trained their models (i.e., local, centralized, and federated) and therefore also highlight the distinct types of configurations that exist for the training of their data.

- Advantage : Proved that FL models significantly outperformed those developed under individual, local training conditions and offered a privacy-compliant solution in adherence to GDPR guidelines afforded through the avoidance of the need for centralized pooling of data; however,
- Disadvantage : Reported that FL produces hybridized models of performance relative to a centralized training model and that the limitation of the authors in their study stemmed from their use of only four EEG sensors to represent wearable hardware; and
- Methododology : Compared the performance of the Federated Random Forest Regressor to those of the Centralized and Individual training conditions as the benchmarks; and
- Results : Demonstrated that performance values for federated models can produce performance results almost equal to those observed when using a centralized model of training and considerably superior to those reported when developing the performance of a single subject training model to classify emotions.
- Dataset : DREAMER data set (EEG and ECG data from 23 subjects).

Author Rui Liu et al. present a Federated Learning framework called FLEEG, which helps improve performance on EEG data from different sources/devices. FLEEG allows different institutions/participants with heterogeneous data to work together to create one single global model by using various local datasets associated with each institution to represent their local data and ensure similarities through the use of global dataset and the permissions associated with it.

- Advantage : Using FLEEG allows institutions to share different types and amounts of data across multiple devices, which overall results in a median classification accuracy improvement of 16.7%, with the largest improvements for small datasets.
- Disadvantage : The hierarchical structure of FLEEG increases the complexity of the model by requiring precise alignment of the local layers of each institution's FLEEG model.
- Methodology : FLEEG is a Hierarchical Personalized Federated Learning method that uses the Local-Global-Local approach for building FLEEG local datasets to allow data to be better combined using multiple different types of EEG devices
- Results : Multiple studies support the claim that the use of FLEEG had consistent improvements in accuracy across all nine public datasets evaluated in the study.
- Dataset : Datasets are comprised of nine public Motor Imagery (MI) datasets including BCI-IV-2a and High-Gamma Dataset (HGD).

In the paper “A Federated Learning Framework for Integrative Multimodal Parkinson’s Disease Biomarker Fusion” [4] by authors Ruchira Pratihari and Ravi Sankar, an innovative framework for Federated Learning (FL) is proposed, which allows for the multimodal fusion of biomarkers (e.g., EEG, fMRI, speech and gait) associated with Parkinson’s Disease (PD). The framework obtains measurements from these heterogeneous data types among different client nodes and fuses them at a central server in order to detect both PD and Freezing of Gait (FoG). This framework builds an effective and secure diagnostic tool that combines multiple clinical indicators, while still protecting the sensitive patient information.

- Advantage : Successfully fuses asynchronous and multimodal biomarkers from various modalities while preserving total data isolation of each data type.

- Disadvantage : The computational overhead associated with the server-side multimodal fusion of the models will increase the time required to aggregate the model.
- Methodology : Local (CNN and LSTM) models, utilizing the Fed-Dyn algorithm and attention-based method for feature fusion.
- Results : Accuracy of the classifications exceeded 85% for detecting PD, and 97.8% for detecting FoG, using the attention feature fusion technique.
- Dataset : PD-DS-I (multimodal) and MOVE (gait and accelerometry) datasets were used.

This article "(Deep Learning Strategies for Achieving Motor Imagery from EEG: State-Of-The-Art Models, Generalization Challenges, And New Directions)" [5] authored Aaqib Raza and Mohd Zuki Yusoff is an evaluation of current deep learning (DL) motor imagery (MI) decoders, with a specific emphasis on eliminating motor imagery (MI) as an interface between people and computers (a true brain computer interface) through real time processing. There are 68 out of a larger population survey that have documented sufficient validation data to support their use in current, state of the art BCI(s). The authors summarize current research towards building federated learning systems and systems that will run on neuromorphic hardware due to the need for low power, privacy-preserving BCI systems as two primary new trends that could have a significant impact on the future evolution of BCI(s).

- Advantage : Provides an extensive benchmarking of the very latest architectures (i.e. SWCNet, TMCNet) as related to MI-based BCIs and systematically types the major challenges within the broad/general area of cross-subject generalization.
- Disadvantage : While it draws conclusions about the literature reviewed by constructing summary tables of each study, it doesn't present a new experimental model or dataset.
- Methododology : This paper is a systematic review of the published literature and a meta-analysis of 68 peer-reviewed BCI articles from 2024 to 2025.
- Results : Identified benchmark accuracies at between 85% -100% across standard

data sets, indicating the clear need for lightweight models.

- Dataset : Literature review and meta-analysis (covering various standard BCI data including BCI IV 2a, PhysioNet, etc.) were used as the data on which to base the conclusions drawn in this paper.

The article "Federated Deep Transfer Learning for EEG decoding Using Multiple BCI Tasks" [6] by Xiaoxi Wei and A. Aldo Faisal introduces a Multi-dataset Federated Separate-Common-Separate Networks (MF-SCSN) framework. It integrates FL (Federated Learning) to preserve user's privacy with deep Learning transfer in order to share knowledge across several BCI (Brain-Machine Interface) tasks. The way this framework works is by allowing the model trained on 'source' task(s) (e.g. hand movement) to help it do well on a 'target' task (e.g. tongue movement) without sharing the original data. It is aimed at overcoming issues of 'negative transfer' when learning from datasets that are not the same.

- Advantages : Maximizing data reusability across various types of protocols creates an opportunity for increasing data availability while maintaining both data-level and parameter level privacy protections.
- Disadvantages : Synchronization of local clients and central server will require three stages to synchronize (complex).
- Methododology : The MF-SCSN Architecture.
- Results : Outperformed baseline decoders by 3% across task during the BEETL (Brain-Esther and Eletromyogram to determine Loop by Beat tracking, Brain-Direct Gradient and Controlled computer-based electroencephalogram) Competition.
- Dataset : NeurIPS 2021 BEETL datasets - Cross task and Cross Dataset EEG.

The article "AGTCNet: A Graph-temporal approach to principled Motor Imagery EEG classification" [7] by Galvin Brice S. Lim et al. presents AGTCNet, an Attentive Graph-temporal Convolutional Network, which combines topographical information about the brain within the classifier – allowing for a better Motor Imagery (MI) EEG classifier. The authors use a graph to model the layout of EEG electrodes in space and employ Graph

Convolutional Attention (GCAT) to jointly learn the spatial relationships and temporal features of the data. By using this "principled" approach the authors maintain biological relevance while also providing computational efficiency for real-time BCI systems.

- Advantages : Substantial improvements in effectiveness: Decrease model size by 49.87% and increase inference speed by 64.65% over traditional CNNs.
- Disadvantages : The assumption of regularity in the graph structure may not apply well to other non-regular or sparsely populated headset configurations.
- Methodology : Integrated graph-temporal convolutional networks with GCAT.
- Results : New state of the art performance on BCICIV2A and PhysioNet datasets.
- Dataset : BCI Competition IV 2a and PhysioNet motor imagery datasets.

Ce Ju and co-authors' paper titled Federated Transfer Learning for EEG Signal Classification [8] introduces a Federated Transfer Learning (FTL) architecture to address the challenge of small sample sizes in Electroencephalography (EEG) and allow for the collaborative use of multi-subject data. The approach is based upon the use of covariance matrices from single trial EEG recordings for each individual subject to extract common discriminative features that can be identified among different individuals by applying domain adaptation without requiring the exchange of raw neural signals between subjects. The FTL framework was developed with the goal of maintaining the privacy of brain data while also increasing the robustness of Deep Learning Decoders for use on EEG signals.

- Advantages : The proposed FTL method provides the ability to close the data gap in Brain-Computer Interfaces (BCIs) with greater accuracy than any available state-of-the-art central model without requiring the exchange of raw neural signals.
- Disadvantages : The use of manifold reduction techniques and covariance matrices is computationally intensive and may not be appropriate when working with high-density EEG systems.
- Methodology : Combine manifold reduction and tangent projection to a given vector space with a federated layer defined using the FedAvg algorithm.
- Results : Increased the accuracy of subject-specific EEG signal classification by 6% and the improved accuracy of adaptive subject-specific network based on the same categorization by 2% compared to benchmark data.

- Dataset : The data set used for evaluation was the PhysioNet 2-class motor imagery data set.

Tianwang Jia et al.'s paper titled "Federated Motor Imagery Classification for Privacy-Preserving Brain-Computer Interfaces" (2020) presents FedBS, a federated framework that addresses Non-IID data and its associated challenges amongst different BCI users. In order to account for individual differences among users, the authors propose a method of local batch-specific normalization and utilize a Sharpness-Aware Minimization (SAM) optimizer to facilitate global generalization of the federated model to unseen subjects. This dual approach will allow for the creation of federated models that are as robust as or more robust than centralized models.

- Advantage : In most cases, FL models outperform equivalent models trained using centralized methods demonstrating that FL can generalize at a level that is superior to privacy-preserving centralized learning.
- Disadvantage : Additional gradient computations are required for SAM as part of the optimization process resulting in additional local training time and power usage.
- Methodology : The FedBS framework uses local batch-specific normalization and a Sharpness-Aware Minimization (SAM) optimizer.
- Result : The performance of FedBS was superior to 6 state-of-the-art FL methods across 3 separate MI datasets.
- Datasets : the BCI Competition IV 2a, 2b, and the PhysioNet datasets.

Kairui Hu et al.'s paper titled 'Horizontal Federated Learning For Brain-Computer Interface' [10] presents a horizontal federated learning scheme to address security concerns associated with transmitting unprocessed brain signals and offers a solution to this problem through the use of a federated deep convolutional neural network (Fed-DCN). A federated deep convolutional neural network is created by collaborating between multiple clients to build a deep learning model while allowing them to only share model weights (i.e., share their representation of all their layers), as opposed to transmitting all of their raw data back to a central server, which would otherwise expose their neurological data to potential cybersecurity risks if there were to be a data breach.

- Advantage : Provides a secure decentralized training paradigm to maintain locality of data, while increasing performance in classifying subjects' brain signals (as opposed to simply training and classifying an individual subject's brain signal).
- Disadvantage : Limited to a single training set, and therefore does not fully address the concerns surrounding negative transfer resulting from high inter-subject variability.
- Methodology : Application of horizontal federated learning to DCN and use of federated averaging (FedAvg) algorithm.
- Result : Consistently and significantly outperform subject-dependent traditional non-federated DCN models.
- Dataset : BCI 2014-001 motor imagery dataset (9 subjects, 4 tasks).

Rao Wei is an author of "Improving Generalization in Federated Learning for Steady-State Visual Evoked Potential Classification" [11] which focuses on classifying Steady-State Visual Evoked Potentials (SSVEP) using a federated framework dubbed FedGF. The framework implements Data-Free Knowledge Distillation (DFKD) which does not require sharing raw data in order to develop generalizable models through a central server generating global features to support local model training. The study then extends its theoretical findings into a practical application through the creation of a BCI model to control the electrical stimulation of a soft gripper to perform robotic tasks.

- Advantages : By applying the FL-trained model to a functional robotic control system, there is a direct correlation between theoretical findings and functional implementation.
- Disadvantages : DFKD is dependent on multiple rounds of complex communication between the generator and clients, thus potentially contributing to increased latency.
- Methodology : Federated Framework (FedGF) using Data-Free Knowledge Distillation (DFKD) and Global Feature Generation.
- Results : 0.65% improvement in accuracy over alternative federated learning methods and also successful application to controlling a robotic system.
- Datasets : Utilizes two public SSVEP datasets and one private SSVEP clinical

dataset.

The article "mixEEG: Enhancing EEG Federated Learning for Cross-subject EEG Classification with Tailored mixup" [12] by Xuan-Hao Liu and his colleagues presents a novel technique named mixEEG, which enhances the cross-subject classification of EEG signals in the context of Federated Learning. This approach makes use of tailored mixup techniques, specifically Channel and Frequency Mixup, to account for the differing spatiotemporal characteristics of EEG data. By implementing the mixing of unlabeled averaged data obtained from subjects that have not previously contributed to the study, the authors are able to improve upon the performance of the Federated Learning model when classifying data from new participants, even though these participants may be located in distributed locations.

- Advantages : This approach provides an alternative means of addressing the variability that is inevitable when working with EEG data by implementing frequency and channel interpolation for either model transfer, thereby enabling more accurate classification of EEG signals that originate from different sources.
- Disadvantages : The implementation of sharing averaged data between participants also represents a small theoretical privacy concern, compared to traditional Federated Learning protocols that only share model weights with other participants.
- Methodology : mixEEG framework with tailored mixup (Channel/Frequency); semi-supervised domain adaptation.
- Results : Improved cross-subject performance; specifically when detecting seizures, as well as in recognizing emotion.
- Datasets : CHB-MIT (seizure detection) and SEED (emotion recognition).

Zhou, Haimudula, and Tang (2016) introduced "Dual-Branch Convolution Network With Efficient Channel Attention for EEG-Based Motor Imagery Classification" [13] describes a new type of network called DB-ATCNet (i.e., a dual-branch attention temporal convolution network) meant to improve brain signal decoding. This architecture contains two main modules: ADBC (i.e., attention dual-branch convolution) and ATC (i.e., attention temporal convolution). The two modules interact with each other in such a way that enables

them to more effectively capture spatial and temporal properties of channel data from the brain. Efficient channel attention is integrated into this methodology, allowing for selective processing of the most useful EEG channels as determined by the attention mechanism, which contributes to greater classification performance with a fewer number of parameters.

- Advantages : Compared to standard deep networks, the use of this architecture improves classification performance while keeping the number of parameters lower indicates it could potentially be implemented for low-power, edge computing applications.
- Disadvantages : The dual-branch nature of this network means that it may continue to have difficulty generalizing when placed into a highly noisy and uncontrolled EEG environment.
- Methodology : Utilize dual-branch convolutional architecture (ADBC & ATC modules) + mixed channel attention.
- Results : Compared to existing models (e.g., EEGNet, DeepConvNet), this architecture produced significantly more accurate results (i.e. higher accuracy + Kappa values).
- Datasets : Evaluated on BCI Competition IV-2a & IV-2b datasets.

In their paper "A Multi-Domain Convolutional Neural Network for EEG-Based Motor Imagery Decoding," Hongyi Zhi et al. introduced the TSFCNet, a multi-domain convolutional neural network model that addresses issues of redundancy inherent in many deep learning architectures. The proposed architecture consists of a MixConv-Residual block that extracts temporal features across multiple scales from a multi-band filtered data set, followed by three independent parallel convolution operations that extract feature representations from spatial, frequency, and time-frequency domains. This use of multiple domains enables strong feature extraction performance without requiring the use of a very deep or complex network architecture.

- Advantage : The use of multi-domain feature representations (spatial, temporal, frequency) allows for reduced overfitting and improved decoding accuracy.
- Disadvantage : Because multiple domains are processed in parallel, the complexity of the feature fusion layer has been greatly increased compared to a more traditional

approach.

- Methodology : Multi-domain TSFCNet utilizing a MixConv-Residual block and three parallel convolution branches for spatial, frequency, and time-frequency domains.
- Results : The proposed method outperformed several current state-of-the-art methods on two publicly available datasets.
- Datasets : BCI Competition IV-2a and BCI Competition IV-2b.

In this paper by Jin et al. in reference [15], proposes a new classification model based on both modeling functional brain connectivity and utilizing graph theory to develop a method for classifying EEG data that supports brain-computer interface (BCI) applications for the purpose of motor imagery. The authors use graph representations of electroencephalograms (EEG) to extract "motifs", or local network structures, from the connectivity graphs to develop a method to use those graph motifs as input to a graph embedding model called Ego-CNNs. By using Ego-CNNs rather than using traditional signal processing techniques, the authors state that their method allows for the possible identification of more critical brain network structures than traditional methods.

- Advantage : Provides a means of achieving high classification accuracy by utilizing the functional relationship between different brain networks instead of focusing on individual channel signals.
- Disadvantage : Computing functional connectivity and motif extractions for every trial may be more computationally expensive and difficult to achieve in real-time BCI applications.
- Methodology : Construct functional connectivity graphs; extract motifs; use Ego-CNN for graph embedding.
- Results : Achieved high accuracies for two-class tasks (up to 96.5%) and for multi-class tasks (90.33%).
- Dataset : Evaluated across 4 separate EEG datasets in order to guarantee robustness of the classification models used in the study.

The research paper "ACNN-STADP: An Adaptive Convolutional Neural Network

With Spatio-Temporal Attention and Dynamic Pathways For Robust EEG-Based Motor Imagery Classification” [16] by Aaqib Raza and Mohd Zuki Yusoff proposes a new adaptable technique to solve inter-subject variability and lack of generalizability in BCI environments. Spatio-temporal attention modules will be used to place focus on important segments of the data during signal processing as well as provide a mechanism for dynamically determining which processing pathways to use at any given time based on the incoming data. Therefore, the adaptive capability of the ACNN-STADP will enhance system reliability among diverse user groups and multiple test sessions.

- Advantage : Addresses non-stationarity of EEG signals; combines adaptively tuned spatio-temporal attention and dynamically selected processing pathways.
- Disadvantage : Increased complexity of developing the dynamic processing pathway; requires advanced optimization capabilities.
- Methodology : Adaptive CNN (ACNN) and spatio-temporal attention and dynamic pathways (STADP).
- Results : Increased robustness and improved classification performance vs. standard static convolutional neural networks.
- Dataset : Evaluated against BCI Comp IV 2a dataset as well as several other standard MI datasets.

2.2 SUMMARY OF LITERATURE SURVEY

The studies combined represent a change in the way we design and construct Brain-Computer Interfaces (BCIs), moving away from a more centralized, opaque and "black-box" model based on Deep Learning, to a model based on decentralized, Privacy-Preserving Federated Learning (FL) and Physio-Informed Graph Architectures (PIGAs). A central premise is that these two newly developed paradigms will alleviate the problems associated with data scarcity and the risks related to privacy by employing decentralized, privacy-preserving frameworks such as FedAvg, Fed-Dyn, and FedBS. These frameworks allow multiple research sites to train highly robust models for assessing stroke, diagnosing Parkinson's Disease, and evaluating motor imagery without sharing sensitive raw electroencephalogram (EEG) data. Concurrent with this shift to decentralized data frameworks is a trend towards improving spatial feature extraction through graph modeling

of the human brain's functional connectivity. Several examples of such Physio-Informed Graph Architecture-based models include AGTCNet, DB-ATCNet and Ego CNNs that employ the scalp surface topology and identify inter-regional brain "motifs" to produce state-of-the-art model performance with significantly reduced computational burden. In addition, current research efforts are focused on achieving cross-subject generalization by employing transfer learning and data augmentation techniques (e.g., mixEEG and Data-Free Knowledge Distillation) to account for the natural variability of brain activity across different people.

Despite laboratory proven high accuracy rates there have been multiple issues preventing the clinically or commercially widespread implementation of these types of technologies. This includes three primary gaps in knowledge regarding:

- The robustness of signal generated in an actual user scenario as opposed to traditional "gold standard" publicly available signal datasets created under highly controlled circumstances. There is very little research on how well low cost, dry electrode wearable BCIs perform with regards to the robustness of their signal generation when subject to a users ability to perform the task (motion artefact) and differences between each user and their ability generate consistent quality signal ("in-the-wild").
- The personalisation vs privacy trade-off with regards to the methods used to ensure that user data remains confidential, and that this use for integration of data into the universal BCI model results in NT/ IID (Non-Independent and Identically Distributed - resulting from uniqueness to individual i.e. differing BCI recording hardware from person to person) must be solved. No research has resulted in the ability to provide an equal balance point between the stability of a global BCI model versus the extreme need for personalisation for each user.
- Explainability (XAI) and clinical trust (in the use of high performing DCIs). Although efforts have been made to incorporate a model for producing "X AI" or "explainable" models; traditional high-performance machine learning models (ML) are mathematically very complex in design. There is also significant lack of research that assists clinicians in accurately mapping the output of the X AI

(Importance Score) to the clinical neurophysiological data associated with the surgical or rehabilitative procedures.

- (Dynamic connectivity adaptation): Most graph models rely on fixed electrode locations as a basis to develop static adjacency matrices. The brain's functional network, on the other hand, is highly dynamic; therefore, there is a need for a method of dynamically learning the topology of a graph to reflect real time changes during the user's mental state or level of attention.
- (End-to-end latency for closed-loop control): For brain-computer interfaces used to control prosthetic devices or robotic hand grippers, every millisecond matters. Current research commonly overlooks the time delays introduced from multiple domains, as well as the time delays associated with communications between federated learning agents (via rounds), that can interfere with the seamless interaction between humans and machines that is required for real-time motor replacements.

Chapter 3

Methodology

3.1 PROJECT DOMAIN

The project involves BCI systems, machine learning and distributed learning frameworks. BCI is technology that allows for direct communication between humans (brain) and other physical devices by interpreting the neural activity of the brain. BCI has applications in the area of assistive technology for people with motor disabilities, neurorehabilitation, gaming and human-computer interaction. Motor imagery is one of the most studied paradigms within BCI. In motor imagery scenarios, subjects imagine performing a physical movement (eg. moving their left or right hand) without physically performing the action. Performing these mental actions generates EEG patterns that can be analyzed using signal processing and machine learning to classify the intended motor action.

The challenge of performing motor imagery classification using EEG data has many inherent difficulties due to inherent characteristics of the neural signals, including low signal to noise ratio, high variability over time and significant differences in brain activity patterns across users. As such, the current state-of-the-art methods using traditional machine learning approaches do not generalize well across users due to the differences in EEG patterns generated by different users. The present research into BCI uses advanced computational methods like deep learning architectures, geometric feature representations, and federated learning frameworks to address various challenges in BCI research. Deep neural networks have been used for learning very complex temporal and spatial patterns from EEG signals. There are also more mathematically meaningful representations of covariance structures found in EEG data that may be captured through geometric methods such as Riemannian manifolds.

There have also been privacy concerns related to EEG data and data is typically collected from many separate sources, and thus the use of federated learning as a means of collaboratively constructing models from the separate data sources without the need to share raw data. For EEG-based BCI systems, this method will provide a means to better understand neural networks due to the highly sensitive nature of the data collected and the data being collected from multiple users/institutions. The overall scope of this project can be described generally as an EEG-based classification of motor imagery using complex machine learning methods in a federated learning framework. The objective of this research project is to improve the classification performance of EEG-based motor imagery through addressing subject variability issues, privacy protection and distributed model training.

3.2 DATASET DESCRIPTION

3.2.1 BCI COMPETITION IV DATASET 2A

Motor imagery classification algorithms are often evaluated with the BCI Competition IV Dataset 2a, a widely accepted benchmark dataset used for assessing EEG-based Brain-Computer Interface (BCI) systems. The dataset consists of EEG recordings from nine healthy participants performing four different motor imagery tasks (left hand, right hand, both feet, and tongue) that involve the imagination of movements. All subjects performed all tasks over the course of two separate sessions (one per day); each subject's recordings contained a total of 288 trials. During each trial, the subject was shown a visual cue that identified which motor imagery task to complete, followed by the subject imagining the corresponding movement for a predetermined amount of time while the EEG signal was being recorded. With this data, the BCI Competition IV Dataset 2a defines a multi-class motor imagery classification problem; following its widespread use, it will continue to serve as a standard method for evaluating algorithms that detect and classify motor imagery-related patterns of brain activity.

Each participant was tested once per session for each of the 4 motor imagery classes; as a result, each participant had 72 trials per motor imagery class per session. In total, this dataset has 1152 trials for each of the 4 classes (288 trials) (and for all subjects (9 subjects)), which provides sufficient data points for evaluating multi-class (motor

imagery classifications). Due to the presence of multiple subjects and sessions, the dataset also enables research on cross-subject generalization and subject variability in EEG signals.

3.2.2 BCI COMPETITION IV DATASET 2B

Dataset 2b of BCI Competition IV is designed to assess classification algorithms based on 2 classes of mental imagery. Unlike dataset 2a, which consists of 4 classes of mental imagery, dataset 2b only evaluates the different mental images of your left hand and right hand. This dataset contains 9 sets of EEG recordings, with each set producing one set of 3 bipolar EEG measurements made at the locations of the 3 bipolar EEG channels so that one channel records activity in the part of the brain that controls hand movement. In addition to the bipolar EEG channels, there are 3 EOG channels that were recording activity associated with eye movements was also recorded.

The samples were taken at 250 Hz, as were the samples in dataset II a. The same subject participated in multiple recordings for training and evaluating BCI systems on separate occasions. At the beginning of each trial, the experimenter presented the subject with a cue indicating whether to think about left-hand movement or right-hand movement; the EEG was recorded during the imagination time period. Dataset II b is especially useful for analysis of two classes of motor imagery classification (i.e., separate mental image of moving left hand versus separate mental image of moving right hand). Since less than half as many EEG channels and only two classes = and consequently = are used, dataset II b is typically used for evaluating simple BCIs or other BCI methods using low-cost components.

Both Dataset 2a and Dataset 2b have become standard benchmarks in BCI research, enabling researchers to compare different signal processing and machine learning techniques for motor imagery classification.

3.3 EEG PREPROCESSING

EEG data contains inherent noise and is easily corrupted by non-brain related

artifacts, such as those created by eye blinks, muscle movements and surrounding environment. For this reason, the data undergo preprocessing in order to improve the data quality and identify the specific neural signals associated with motor imagery tasks.

3.3.1 BANDPASS FILTERING

Specific neural oscillations will be experiencing motor imagery activity in certain frequencies of EEG signals, mainly in the mu band (8-13 Hz) and beta band (18-26 Hz). To gather those EEG signals by a bandpass filter, the EEG signal has been filtered within the bands. Filtering will allow low-frequency drift to be eliminated and high-frequency noise to be eliminated; therefore, the features will represent meaningful brain wave patterns.

3.3.2 TRIAL SEGMENTATION

To obtain input samples for feature extraction and model training, individual motor imagery events are segmented into trials. Each trial is derived from continuous EEG recordings taken during the time windows when subjects perform various motor imagery tasks.

3.4 ATTENTION-BASED TEMPORAL CONVOLUTIONAL NETWORK (ATCNet)

ATCNet (Attention Temporal Convolution Networks) is a type of deep learning network that has been developed to assist researchers in analyzing and identifying samples taken from the EEG (electroencephalography) use case. ATCNet is an effective model for identifying motor imagery-based classifications for BCI systems through capturing both temporal dynamics and discriminative characteristics of BCI signals at the same time. The characteristics of EEG signals are typically shown as being very complex – they are often non-stationary and full of noise. Most traditional methods utilize predetermined feature extraction methods, which do not necessarily contain the correct set of temporal dependencies for an individual's neural signal. ATCNet overcomes this limitation through the use of a combination of techniques together with the temporal convolution network technique coupled with attention mechanisms, enabling the network to automatically learn

temporal features from EEG samples without predetermined features being specified. The overall architecture of the ATCNet network includes several major components that allow independent components to be used in a coordinated manner to extract useful information from EEG signals.

3.4.1 TEMPORAL CONVOLUTION MODULE

The ATCNet framework is built on a basis of a temporal convolution module. Temporal convolutional layers convolve along the timeline of the EEG signal with convolutional filters that capture the temporal characteristics of the signal, specifically the rhythmic patterns like Motor imagery oscillations. Changes in specific EEG frequency are present with motor imagery activity; more specifically in the case of motor imagery the mu rhythm (8–12 Hz) and beta rhythm (18–26 Hz) frequency bands. The temporal convolutional layers learn to detect these patterns by learning filters that recognize salient temporal features of the signal from the EEG data.

Temporal convolutional networks are very different from recurrent neural networks as they do not suffer the same problems during gradient training because they are able to process sequences in parallel and maintain the stability of the gradients as they pass through training. Thus, they are ideal for modeling long-term temporal dependencies in EEG data.

3.4.2 SLIDING WINDOW FEATURE EXTRACTION

ATCNet uses a sliding window approach to evaluate the temporal variability inherent in the EEG signal. The EEG trial is divided into multiple overlapping temporal segments that are processed independently via each of the convolutional layers; thereby allowing the network to examine many different segments of the EEG signal, as well as determine which temporally distinctive areas correspond to motor imagery activity.

3.4.3 ATTENTION MECHANISM

The attention mechanism is another crucial aspect of ATCNet because it allows the model to focus on the relevant temporal features extracted from the convolutional layers. The EEG signal used for motor imagery tasks does not always have equal contributions to

the classification of an event. To account for this, the attention mechanism is able to provide relevance weightings to all of the temporal features extracted by the convolutional layers (i.e., all of the relevant features of a particular EEG signal), providing the network with a greater opportunity to determine what patterns of data are important to the classification process and to reduce the influence of other irrelevant/noisy types of data on the network's performance. Through its ability to selectively focus on patterned features, the attention mechanism positively impacts the performance of the overall model.

3.4.4 FEATURE AGGREGATION AND CLASSIFICATION

The aggregation of the convolutional and attention features occurs at the compact-level representation of each EEG trial. The compact-level representation is then used as input to the fully connected layers, which provide the final classification of the motor imagery task. The softmax activation function is used in the last layer to calculate the probability score of each category of motor imagery.

3.4.5 ADVANTAGES OF ATCNET FOR EEG-BASED BCI

ATCNet's application to motor imagery classification via EEG has many benefits:

- 1) The temporal dependencies of EEG signals are captured very effectively.
- 2) The use of an attention mechanism will help provide improved robustness against noise.
- 3) The temporal convolutional layers allow for the learning of features in an efficient way compared to more traditional recurrent architectures.
- 4) The structure of ATCNet has been demonstrated to be capable of handling multi-channel EEG signals and multi-class motor imagery tasks.

Given the aforementioned characteristics, ATCNet has achieved state-of-the-art accuracy on the BCI Competition IV Dataset 2a; thus, acting as a suitable backbone to EEG-based motor imagery classification systems.

3.5 RIEMANNIAN MANIFOLD REPRESENTATION FOR EEG SIGNALS

In EEG-based Brain–Computer Interface (BCI) systems, the ability to extract meaningful features from the neural signals is critical for accurately classifying motor imagery tasks. Traditional machine learning techniques use Euclidean spaces to represent EEG feature data in terms of the usual vector operations of averaging and distance calculation. This may be problematic for some EEG feature representations, particularly covariance matrices, which do not naturally map to Euclidean space, rather they are located on a curved geometric structure, known as a Riemannian manifold. A Riemannian manifold is defined mathematically as a space in which each point is characterized by a smoothly varying geometric structure and defines distances and angles in a local manner. For this reason, from an EEG signal analysis perspective, covariance matrices generated from EEG data are classified as members of the set of Symmetric Positive Definite (SPD) matrices, which form a Riemannian manifold and not a flat Euclidean space.

3.5.1 COVARIANCE REPRESENTATION OF EEG SIGNALS

EEG Signals which have been recorded from many electrodes deliver significant spatial information regarding brain activity. To capture the relationships between EEG channels, covariance matrices are computed for each EEG trial. Let an EEG trial be represented as a signal matrix:

$$X \in \mathbb{R}^{C \times T} \quad (1)$$

Where,

C = number of EEG channels

T = number of time samples

The covariance matrix is calculated as:

$$C = \left(\frac{1}{T}\right) X X^T \quad (2)$$

The covariance matrix conveys the correlation structure between the EEG channels while performing a motor imagery task. Two key characteristics of the covariance matrix are:

- 1) They are symmetric.
- 2) All of the eigenvalues are positive.

By possessing these two characteristics, covariance matrices belong to the class of Symmetric Positive Definite (SPD) matrices forming a curved geometric manifold.

3.5.2 LIMITATIONS OF EUCLIDEAN METHODS

Applying the standard Euclidean operations directly to a covariance matrix, you may distort the way geometric structures of a covariance matrix are represented. For example, computing the Euclidean distance between two covariance matrices as,

$$\|C_1 - C_2\| \quad (3)$$

This distance does not reflect the intrinsic geometry underlying a SPD matrix. Therefore, if you perform such an operation as described previously you may create inaccurate similarity measures between EEG trials.

3.5.3 RIEMANNIAN DISTANCE BETWEEN COVARIANCE MATRICES

The use of Riemannian geometry provides us a mathematically sound manner to measure the distance between covariance matrices by taking into account the curvature of the manifold spanned by the SPD matrices. So instead of measuring distances in a straight line, we are measuring distances in the manner of the shortest distance (geodesics) along the curvature of the SPD manifold. One of the commonly used Riemannian distance metrics for measuring the distance between two covariance matrices is given by:

$$d(C^1, C^2) = \left\| \log \left(C^{1(-\frac{1}{2})} C^2 C^{1(-\frac{1}{2})} \right) \right\| \quad (4)$$

Where,

C_1 and C_2 = covariance matrices

$\log(.)$ = matrix logarithm operation

$\|.\|$ = Frobenius norm

Essentially, this distance metric enables us to protect the geometric structure of the

covariance matrices and serves as a more accurate form of quantifying the similarity (distance) between EEG trials.

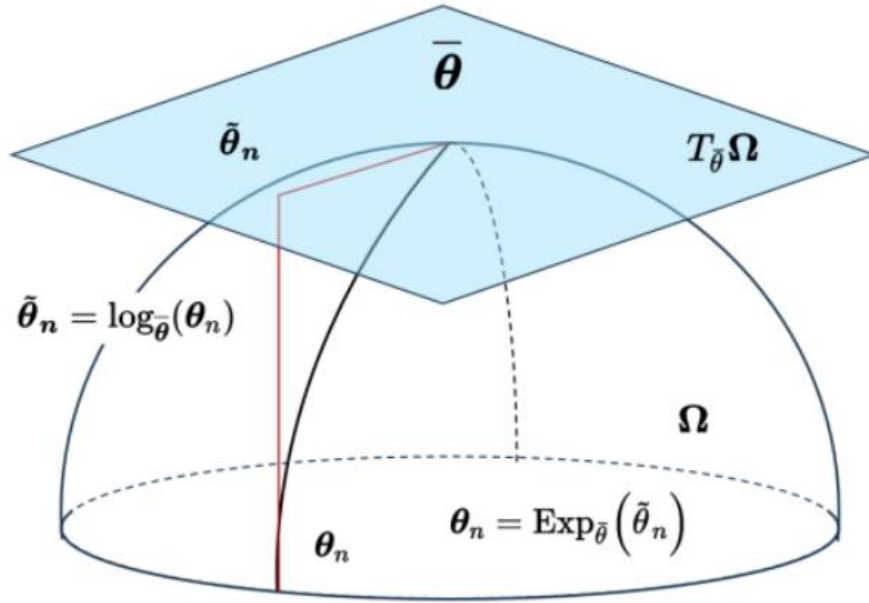


Figure 3.1 - Corresponding local tangent space at a reference point

3.5.4 TANGENT SPACE MAPPING

Machine Learning algorithms mainly use Euclidean spaces for distance calculations despite Riemannian geometry being useful in making distance calculations on curved manifolds. By using the Logarithm of the Cholesky of C (a covariance matrix) and C_{ref} (a reference matrix) to find their distance from tangent space, it provides for more linearity. Tangent space at a reference point is shown in Figure 3.1. The mapping to tangent space is performed using the matrix logarithm:

$$T = \log \left(C_{\text{ref}}^{-\frac{1}{2}} C C_{\text{ref}}^{-\frac{1}{2}} \right) \quad (5)$$

Where,

C = covariance matrix of a trial

C_{ref} = reference covariance matrix (usually the geometric mean of all covariance matrices)

Once the tangent space has been reached through the matrix logarithm, the T matrix can

be normalized and flattened so that it can serve as input to ML/DL algorithms like convolutional networks.

3.5.5 ADVANTAGES OF RIEMANNIAN MANIFOLD METHODS

In EEG signal analysis, Riemannian manifold-based methods offer various advantages:

- 1) They maintain the intrinsic geometric characteristics of covariance matrices.
- 2) They provide an ability to measure distance between EEG trials meaningfully.
- 3) They improve resistance to noise and signal variability.
- 4) They allow effective feature extraction for classifying motor imagery.

Because of these advantages, Riemannian geometry is now one of the primary techniques used in contemporary BCI research based on EEGs, particularly in connection with covariance-based feature representations and multi-subject EEG datasets.

3.6 FEDERATED LEARNING FRAMEWORK

To mitigate the issues around the use of an experimental program with regard to privacy and distributed data availability, a federated learning approach is utilized to train the model. Under this approach:

- 1) A client is conceptualized as an independent subject.
- 2) A central server performs the aggregation of all clients' models.
- 3) All raw EEG data will remain stored locally on each client.

The federated training process is implemented following an iterative communication protocol. The federated training procedure is performed as follows:

- 1) The central server initializes the parameters of the global model.
- 2) The global model is distributed to each of the participating clients.
- 3) Each client performs local training of the model using the EEG dataset which belongs to that client.
- 4) Once local training is completed, client-specific model updates are then sent back to the central server.

- 5) The central server aggregates the model updates sent by each client to produce a new global model.

The aggregation step uses the Federated Averaging (FedAvg) algorithm to calculate the new global model:

$$w_{\text{global}} = \sum \left(\frac{n_k}{n} \right) w_k \quad (6)$$

Where,

w_k represents the model parameters from client k

n_k represents the number of samples at client k

n represents the total number of samples across all clients.

The cycle will repeat for several communication rounds until the model convergence occurs.

3.7 PERSONALIZED FEDERATED LEARNING STRATEGY

EEG signals are unique to each subject, which makes a single global model ineffective. As a solution to overcome this limitation, a two-phase training method is proposed that involves model personalization.

3.7.1 PHASE 1: GLOBAL FEDERATED TRAINING

The purpose of Phase 1 is to perform global training of the shared parts of the neural network model. In this phase, all clients train a shared model collaboratively using federated learning. This allows for the model to learn general patterns in EEG signals for all subjects.

3.7.2 PHASE 2: LOCAL PERSONALIZATION

Once Phase 1 is completed, each client will perform additional local training of the model using only their own data. During Phase 2, each client will locally adapt certain portions of the model (i.e., modification of spatial layers and classification head to be client-specific) while leaving most of the model unchanged among clients.

The combination of these phases provides the ability for the model to recognize spatially dependent EEG signal features based on individual subjects, but still allows for the use of knowledge shared between clients. The improvement in classification accuracy as a result of this personalization will significantly help adjust the model to each client's brain activity pattern.

3.8 AGGREGATION METHODOLOGY

The purpose of federated learning is to allow a large number of clients from different locations to collaborate on building a single model without sending their raw data to a centralized location. In the study we proposed, each subject is considered an individual client, and each uses only their own EEG dataset to train a local model prior to sending the parameters of that model to a central server to aggregate and update the global model.

The aggregation method selected by the server is important in federated learning because EEG datasets are not normally distributed (i.e., they follow a non-Identically Distributed or non-IID pattern). Variations in electrical activity between individual brains, differences in how electrodes are placed and how signals are measured can produce different models across clients; therefore, an effective aggregation method must be incorporated into the training of the global model. Thus, this section of the report discusses the FedAdapt aggregation method that we propose, in addition to many of the more common methods that have been employed in the ablation study as a point of comparison.

3.8.1 PROPOSED METHOD: FEDADAPT

The FedAdapt aggregation technique is designed to foster robust protected education with significant gene distortion among patients from covariate distributions when used in collective (e.g., federated/Eeg) learning. Key to the FedAdapt approach is an adaptive client-driven rating of contributions of client-initiated contributions in terms of their client-specific contribution, while also making it possible for personalized layers (e.g., shared transfer function layers and/or personalized filtering layer) to be retained across multiple clients. The global model parameters are computed using a weighted aggregation scheme:

$$w_{\{global\}}^{\{t+1\}} = \sum_{k=1}^K \frac{n_k}{n} w_k^{\{t\}} \quad (7)$$

Where,

$(w_{\{global\}}^{\{t+1\}})$ = updated global model parameters at round (t+1)

$(w_k^{\{t\}})$ = parameters obtained from client (k) after local training

(n_k) = number of samples at client (k)

$(n = \sum_{k=1}^K n_k)$ = total number of samples across all clients

(K) = total number of participating clients.

In a federated learning context with FedAdapt, in each round of communication, the client (k) will use its local dataset (k) when training the shared layers of the model and will produce its own level of parameter (w_k) based on training on the locally available data. The resulting w_k 's would typically be exchanged with the central server for aggregation. While both global and local educational phases to the training of the FedAdapt network follow traditional federated learning, the "global federated training phase" of the FedAdapt training process uses a one-phase educational process to accomplish this functional agreement with FedAdapt and the "local personalized phase" that updates shared network layers (e.g., spatial filtering and classification) by localizing the personalization layer at the client level on a given communication round-by-round. FedAdapt will learn to generalize the EEG signal from a global perspective, while allowing it to be personalized to the individual patient.

3.8.2 ABLATION AGGREGATION METHODS

In order to assess how well the proposed Fed-Aaptation strategy works, many standard federation aggregation techniques are applied as comparisons in the ablation (federated) study.

1) Federated Averaging (FedAvg)

FedVg is an effective method (except for the fact that it has difficulty working when there is a lot of information from different places). It also permits instability due to significant differences in the information provided. In FedAvg,

the server aggregates client model parameters by computing a weighted average based on the number of local samples at each client. The aggregation rule is given by:

$$w_{\{global\}}^{\{t+1\}} = \sum_{k=1}^K \frac{n_k}{n} w_k^{\{t\}} \quad (8)$$

where the notation follows the same definition as above FedAvg is simple and efficient; however, it may suffer from instability when client data distributions are highly heterogeneous.

2) Federated Balanced Sampling (FedBS)

The benefit of Federated Balanced Sampling (FedBS) is that it attempts to balance each client to represent an average across all clients, rather than simply using each client's data to provide a representative sampling. By adjusting the sampling, this provides the ability for all clients to have equal contributions towards global updates by preventing any individual from being dominant based only on the size of the contribution. The aggregation can be represented as:

$$w_{\{global\}}^{\{t+1\}} = \sum_{k=1}^K \alpha_k w_k^{\{t\}} \quad (9)$$

where (α_k) represents a balancing weight assigned to client (k) . This approach reduces the dominance of clients with large datasets.

3) Federated Normalized Averaging (FedNova)

FedNova solves the problem of having differing objective functions due to each client having their own number of local training steps. Rather than averaging parameters directly, FedNova normalizes the parameter updates according to how many local optimization steps have been used per update. The update rule can be expressed as:

$$w_{\{global\}}^{\{t+1\}} = w_{\{global\}}^{\{t\}} + \sum_{k=1}^K \left(\frac{n_k}{n} \Delta_k \right) \quad (10)$$

Where, Δ_k represents the normalized update from client (k). FedNova improves stability when clients perform different numbers of local updates.

4) Momentum-Based Federated Optimization

Momentum-based federated learning involves an added momentum term which can help to stabilize global updates, and consequently accelerate the speed of convergence. The server keeps a velocity vector of past global updates to build up a store of knowledge on how much velocity in past updates. The update equations are:

$$v_{\{t+1\}} = \beta v_t + \sum x_{\{k=1\}}^{\{K\} \setminus \{n_k\}} \frac{1}{n_k} (w_k^t - w_{\{global\}}^t) \quad (11)$$

$$w_{\{global\}}^{\{t+1\}} = w_{\{global\}}^{\{t\}} + v_{\{t+1\}} \quad (12)$$

Where,

(v_t) = momentum term

(β) = momentum coefficient.

This approach helps smooth noisy updates from clients.

5) SCAFFOLD

The SCAFFOLD technique introduces the control variates which assists in minimizing the client drift since many clients and servers have non-IID data distributions. Therefore, each client has control variables that complete the local updates during the training process and subsequently help to correct them. The update for client (k) can be expressed as:

$$w_k^{\{t+1\}} = w_k^{\{t\}} - \eta \left(\nabla f_k(w_k^{\{t\}}) - c_k + c \right) \quad (13)$$

Where,

(η) = learning rate

(c_k) = client control variate

(c) = global control variate.

This correction mechanism helps maintain consistency between client and server optimization trajectories.

6) Federated Proximal Optimization (FedProx)

FedProx is the algorithm that is similar to FedAvg, but it has added an additional term (the proximal regularization term) into the local objective function. The proximal regularization term will help to keep the updates of the local model as close to those of the global model. The local optimization objective becomes,

$$\min_w f_k(w) + \frac{\mu}{2} \|w - w_{\{global\}}\|^2 \quad (14)$$

Where,

$f_k(w)$ = local loss function for client (k)

μ = regularization parameter controlling the strength of the proximal term.

FedProx improves stability in highly heterogeneous federated environments.

The important components of a federated learning system are the aggregation strategy and the proposed method FedAdapt. The proposed aggregation strategy allows global shared knowledge to be collected and shared, as well as allowing clients to be adjusted specifically to their own subject data. In order to evaluate if this new approach has any benefit over existing techniques, an ablation study comparing various aggregation algorithms will be completed, including FedAvg, FedBS, FedNova, momentum-based optimization, SCAFFOLD and FedProx. The results from comparing these algorithms will provide the information needed to evaluate the ability of the new aggregation strategy in accommodating heterogeneous datasets, and more specifically, EEG datasets for motor imagery classification.

3.9 OVERALL ARCHITECTURE

The framework that has been presented is a new model which will carry out the

task of motor imagery classification using EEG data collected by a federated learning framework, utilising Riemannian manifold features along with an attention-based temporal convolutional neural network architecture (ATCNet), and its workflow is shown in Figure 3.2. The whole system is designed to allow multiple participants to share in training their own example data sets while maintaining their privacy and variability inherent in their respective EEG samples. The processing pipeline has been broken down into a sequence of ‘layers’ consisting of EEG capture, filtered raw data reception, covariance feature extraction, Riemannian manifold mapping, development of high-level deep neural features, a federated aggregation process for developing participant specific classifiers, and a process for creating participant specific training data via redundantly linked classifiers. The combination of all these layers will produce a series of outputs which can be used to classify participant movements from recorded EEG data associated with previous classifications made by each of the distinct participants involved in this study.

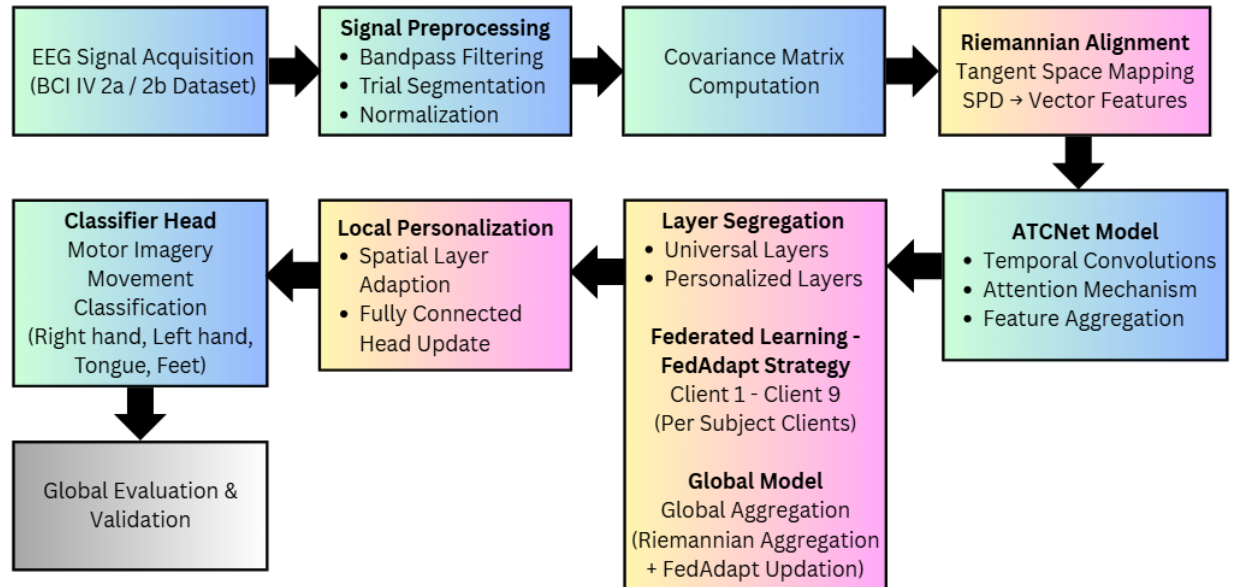


Figure 3.2- Overall Workflow Diagram of the Proposed Model

3.9.1 EEG SIGNAL ACQUISITION

EEG data capture is initiated via placement of a number of electrodes on the scalp according to an internationally recognised standard; electrodes placed over a number of regions should yield a sufficient quantity of raw brain waveforms that can be used to

classify movement in three dimensions while completing various motor imagery tasks involving movement of their left and right upper limbs, lower limbs and/or mouthing off movements associated with their tongue during execution of the respective tasks. A total of 128 channels of EEG data generated from the specified number of participants will represent the neural activity associated with motor imagery testing for each motor imagery type potentially produced during performance of the above tasks.

3.9.2 SIGNAL PREPROCESSING

The EEG recordings are first filtered to remove noise and artifacts, and to keep the relevant frequencies associated with Motor Imagery. In particular, the mu and beta bands are preserved since they are used to represent brain activity associated with movement. Each filtered EEG signal is segmented into separate trials that correspond to individual Motor Imagery events; these trials will be utilized as input samples for feature extraction and model training.

3.9.3 COVARIANCE-BASED FEATURE EXTRACTION

Once the trials are created from the filtered EEG signals, covariance matrices are generated for each trial to characterize the spatial relationships among EEG channels. The covariance matrices contain information about the degree of correlation between EEG electrode locations and help provide insight into how the brain is functioning during a Motor Imagery task. The covariance matrices have a symmetric positive definite structure and maintain a described series of spatial dependencies between channels.

3.9.4 RIEMANNIAN MANIFOLD TRANSFORMATION

Because covariance matrices exist in an alternate geometric space that is not Euclidean but is referred to as the Riemannian manifold of symmetric positive definite matrices, using standard machine learning approaches directly on these covariance matrices may result in poor representation of the data. To combat this limitation, the covariance matrices are mapped to a tangent space using Riemannian geometric procedure. This tangent mapping allows for the covariance matrices to be expressed as vector forms without losing their unique geometrical characteristics for subsequent use with deep

learning algorithms.

3.9.5 DEEP FEATURE LEARNING USING ATCNET

The transformed images are then input into the Attention Temporal Convolutional Network (ATCNet) to learn features and classify motor imagery. Convolutional layers applied to EEG signals along the time dimension allow ATCNet to extract temporal patterns. The architecture of ATCNet also includes an attention mechanism that assigns degree to each temporal segment, so the model emphasizes only relevant time segments when making predictions. ATCNet combines all extracted features to produce predictions for motor imagery classes.

3.9.6 FEDERATED LEARNING FRAMEWORK

Rather than train a model using a centralised dataset of EEG results, this implementation is based on the use of a Federated Learning framework. In the Federated Learning framework, each subject functions as an independent client with access to their own individual data and locally develops a model which is updated periodically as new data become available to provide the most accurate predictions. Once the clients have finished performing their local training, they communicate their updated parameters to a central server that aggregates the new parameters into a new global model. The proposed Federated Learning approach allows the models to be trained collaboratively while providing privacy and no central storage of raw EEG data at the clients' locations.

3.9.7 PERSONALIZED MODEL ADAPTATION

Given that EEG signals can differ greatly among individuals, it may not be possible for one global model to perform effectively across all users. To address this limitation, the proposed architecture includes a personalization phase after the initial round of global federated training. During this phase, specific layers of the model (i.e., the spatial feature extraction and final classification head layers) will be personalized based on the data of the end user. However, all other layers can be kept as shared elements between users in order to maintain the global representation learned during federated training. This will allow the model to learn about characteristics unique to the individual user while also maintaining

knowledge that has been learned across multiple users.

3.9.8 FINAL CLASSIFICATION OUTPUT

Once an EEG signal has been converted into a set of representation features through feature extraction and has gone through the training process, the output of the classification will be a prediction of which motor imagery class corresponds to each of the different ways an individual imagined moving (i.e., left or right arm) during every trial. The classification output will inform the end user of which movement they are imagining. The combination of utilizing Riemannian manifold representations for representing features, using deep learning for making predictions based on the representation features, and using federated training for preserving privacy and increasing classification performance from multiple users' datasets will allow for an efficient and effective solution to processing EEG data for multi-subjects while preserving users' privacy. A diagram of the overall architecture design for this solution is illustrated in Figure 3.1 and demonstrates how the four components of this solution (i.e., pre-processing, feature extraction, deep learning, and federated training) all interact with each other.

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Chapter 4

Results & Discussions

The main goal of this section is to conduct experimental evaluation of the proposed Federated Riemannian ATCNet framework designed to classify motor imagery. In this section we evaluate the performance of this method in a federated learning context using multi-subject EEG data; as well as the performance of classifying multiple types of motor imagery. The performance metrics that will be used to evaluate the performance of the proposed method include accuracy, precision, recall, and F1-score. The goal of the experiments is to demonstrate how our proposed motor imagery classification system will improve performance by combining three approaches: (i) features represented in a Riemannian manifold, (ii) ATCNet architecture, and (iii) personalisation through federated learning, while at the same time addressing differences in EEG signal variability across subjects.

4.1 COMPUTING ENVIRONMENT

We performed all experiments around this proposed framework utilizing cloud-based computation platforms so that there were enough computation resources available to train deep learning models and simulate a federated learning environment. The primary development and experimental environments for this study were Kaggle Notebooks and Google Colab.

4.1.1 KAGGLE ENVIRONMENT

A cloud-based data science ecosystem by way of accessing GPUs and installing machine learning libraries beforehand; thus; it was mainly performed in the Kaggle notebooks for the creation of the proposed model and the execution of federated learning experiments. Furthermore; because the environment at Kaggle contained GPU assistance for deep learning training, pre-configured machine learning libraries, an efficient means for data storage and management capabilities through its system, it allowed for quick

completion of the training process and experimentation with many different aggregation methods to take place quickly.

4.1.2 GOOGLE COLAB ENVIRONMENT

During development and testing of the framework, google colab also has proved quite useful for prototyping models and running experiments. One benefit of colab is the fact that it is free to use, and provides access to many popular deep learning frameworks and also includes the use of cloud-based GPU resources. Because we had access to both kaggle and google colab, we were able to run extensive debugging and validation on the proposed framework.

4.1.3 SOFTWARE LIBRARIES

The suggested framework was created with standard tools which represent how the scientific community uses machine learning and numerical computing to process and present data, namely:

- 1) The programming language is python
- 2) The neural network's framework is Pytorch
- 3) The numerical computation tool is NumPY
- 4) The scientific computing library is SciPY
- 5) The evaluation metrics package is Sklearn
- 6) The package used for importing data sets is MOABB

All of these give the necessary components to be able to build the Riemannian Feature Extraction Pipeline and ATCNet architecture, as well as allow for the simulation of federated learning.

4.2 EVALUATION METRICS

In order to evaluate the proposed motor imagery classification framework, several standard classification metrics were used as a means of evaluating the framework by assessing how it performed. Classification metrics were used to measure the various criteria

associated with evaluation of model performance and allow assessment of the framework's classification performance.

1) ACCURACY

Accuracy is the measure of the number of correctly classified samples among all the samples in the dataset. Accuracy is among the most commonly used classification performance metrics. The accuracy is defined as:

$$Accuracy = \frac{(TP + TN)}{TP + TN + FP + FN} \quad (15)$$

where:

TP = True Positives

TN = True Negatives

FP = False Positives

FN = False Negatives

Accuracy is a general measure of the performance of the framework's classification performance for all the classes.

2) PRECISION

Precision is the measure of the number of correctly predicted positive samples among all the samples predicted to be positive samples, thus precision is a measure of how reliable positive predictions made by the model are. Precision is defined as:

$$Precision = \frac{TP}{TP + FP} \quad (16)$$

A higher precision value means that there are fewer false positive predictions made by the model.

3) RECALL

Recall (also referred to as sensitivity) is the measure of the number of actual positive samples identified correctly by the model. Recall is defined as:

$$Recall = \frac{TP}{TP + FN} \quad (17)$$

A higher recall value means the model has accurately detected most of the true positive samples.

4) F1-SCORE

The F1-score is a measure of model precision and recall that is based on the concept of harmonic mean. An algorithm that gives us a measure of how well the model performs, based on both false negatives and false positives. The F1-score is defined as:

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (18)$$

The F1-Score is particularly helpful when assessing the performance of an algorithm or model because it uses multiple classes that may have different distributions and provides both overall method performance metrics and class-based performance metrics. The F1-score will be used in the remainder of the chapter to review the overall performance and per class performance of the data for the motor imagery classification task.

4.3 ABALATION STUDY – COMPARATIVE ANALYSIS

An ablation study was carried out to test the new aggregation method. The goal of the ablation study is to see if different methods of aggregating data affect how well the federated motor imagery classification model converges and performs globally. The model used for all experiments in this section was the ATCNet model. To ensure that the comparison among the aggregation methods was fair, all other parameters were kept constant throughout the experiments. Specifically, the federated training for all aggregation methods took place over 30 communication rounds, with 10 local training epochs on each client before sending model updates to the central server. The learning rate used for all of the experiments was 0.001.

The aggregation methods tested within the scope of this study include: FedAvg, FedProx, SCAFFOLD, Momentum-Based Aggregation, FedNova, and FedBS. All of the methods evaluated were applied in the same federated learning environment using the BCI Competition IV Dataset 2a. Finally, the end result after 30 communication rounds is shown in Table 4.1 for each method (aggregate accuracy and loss) of aggregation method used.

Table 4.1 - Comparison of Aggregation Strategies on BCI-IV Dataset 2a

S. No.	Aggregation Strategy	Final Global Loss	Final Global Accuracy
1)	FedAvg	0.6052	0.4703
2)	FedProx	0.6313	0.3918
3)	SCAFFOLD	1.9108	0.2500
4)	MOMENTUM	0.8252	0.2979
5)	FedNova	0.6067	0.4310
6)	FedBS	0.6059	0.4157

According to Table 5, FedAvg achieved the greatest accuracy globally at 47.03% and FedNova had the next highest value of 43.10% but somewhat lower than FedAvg. FedBS also produced results of earth-like quality with an achieved accuracy of about 41.57% overall .The extremely low levels of performance for FedProx (39.18% global accuracy) indicate that the proximal regularization term had little or no success at improving convergence or learning for this data set. Momentum method's performance is very poor when compared to all other methods and is reflected in that it resulted on average only 29.79% accuracy. It is also illustrated in Figure 4.1.

This suggests that there is possible slow convergence while federated learning was taking place. Also, SCAFFOLD learned very poorly with only 25% accuracy showing that it learned nothing through its attempts to optimize during process. In addition, the SCAFFOLD loss values were very erratic/unstable throughout the entire federated learning process as evidenced by the expedience of continuing with audits, which contributed substantially to the unstable environment and difficulty in stabilizing the optimization path.

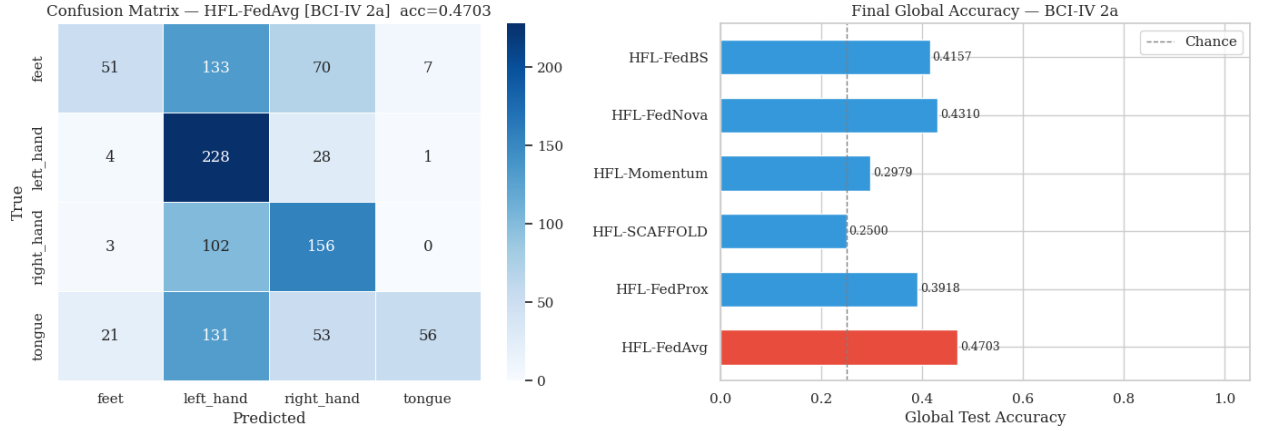


Figure 4.1 - Confusion matrix and Accuracy Bar graph for BCI Competition 2a

The aggregation comparison continued with an ablation study using BCI Competition IV Dataset 2b, which consists of a binary classification task using motor imagery from the right and left hand. Dataset 2b is likely to achieve higher classification accuracy since it has fewer EEG channels and there are only two classes for classification, compared to four in Dataset 2a. The full experimental setup was used for comparison purposes.

Table 4.2 - Aggregation Strategy Comparison on BCI-IV Dataset 2b

S. No.	Aggregation Strategy	Final Global Loss	Final Global Accuracy
1)	FedAvg	0.5574	0.6825
2)	FedProx	0.5630	0.6833
3)	SCAFFOLD	1.0226	0.5000
4)	MOMENTUM	0.5787	0.6672
5)	FedNova	0.5580	0.6733
6)	FedBS	0.5548	0.6879

FedBS performed best achieving 68.79% global accuracy, followed closely by FedProx achieving 68.33% global accuracy and FedAvg achieving 68.25% global accuracy; therefore, these three methods were comparably effective under the federated training setup. Furthermore, FedNova produced competitive results within federated training although it eventually only achieved 67.33%, while the momentum-based method achieved an even lower 66.72% and also exhibited much slower convergence relative to others. The

bar graph is shown in Figure 4.2. Notably, similar to the previous results from Dataset 2a, the SCAFFOLD method had highly variable and unstable performance; it achieved a consistently low 50% accuracy, which indicates random predictions for the binary classification task. In addition, the loss values across trials had large variances during the training phase indicating that optimization was not stabilized for this experimental setup.

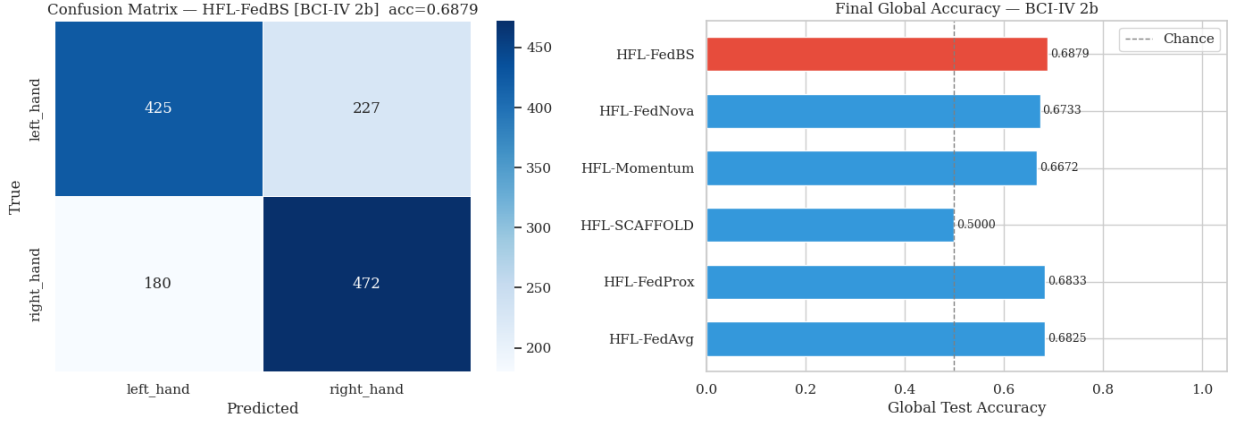


Figure 4.2 - Confusion matrix and Accuracy Bar graph for BCI Competition 2b

4.4 CLASSICAL ATCNET WITH FEDADAPT

To evaluate how effective the proposed FedAdapt training strategy would be, we included a personalization phase after our global federated training. Each client will do more local training to adapt the final global model to their specific EEG data during this phase; this step enables the model to learn unique neural patterns that are not captured in the final global model. This step of the personalization phase is critical for EEG-based Brain–Computer Interface systems since there is a very high level of variability between subjects with respect to their EEG signals. There are often many differences among subjects in the way they have EEG signals when they are active (e.g., brain activity, electrode placement) and in their physical make-up (e.g., gender, age). This means that we are unlikely to ever find a truly global model that can accurately predict all subjects' EEG data because of the wide variety of conditions that exist in we can observe across all subjects. Table 4.3 shows the accuracy of classifying each subject before and after the personalization phase for Dataset 2a.

Table 4.3 - Client Personalization Results on Dataset 2a

Clients	Accuracy before Personalization	Accuracy after Personalization	Improvement
Client 0	0.5948	0.8103	+0.2155
Client 1	0.3672	0.4828	+0.2156
Client 2	0.7500	0.8621	+0.1121
Client 3	0.3448	0.5517	+0.2069
Client 4	0.3621	0.4052	+0.0431
Client 5	0.3534	0.5172	+0.1638
Client 6	0.5259	0.8362	+0.3103
Client 7	0.6983	0.8276	+0.1293
Client 8	0.5776	0.8534	+0.2759

Overall, these findings illustrate that personalizing a model has a significant impact on its accuracy at classifying data into categories for all individuals. Some of the individual personalizations produce improvements in overall accuracy of 20–30%, which reveals that tailoring to an individual's specific (i.e., subject-specific) characteristics will increase the usefulness of the global federated model. Client 6 experienced the most significant improvement following a personalization of approximately 31.03%; and Clients 0, 1, 3, and 8 all experienced improvements exceeding 20%. In addition, the improvements resulting from personalization demonstrate that the global federated model will provide general representations of motor imagery to each individual; however, once it is personalized, the model will be able to represent individuals' unique characteristics. Subject performance before and after personalizing models on Dataset 2b is provided in Table 4.4.

Table 4.4 - Client Personalization Results on Dataset 2b in PHASE 2

Clients	Accuracy before Personalization	Accuracy after Personalization	Improvement
Client 0	0.6597	0.7083	+0.0486

Client 1	0.5000	0.4779	-0.0221
Client 2	0.5000	0.4931	-0.0069
Client 3	0.8514	0.8851	+0.0338
Client 4	0.7095	0.7027	-0.0068
Client 5	0.6042	0.6111	+0.0069
Client 6	0.7569	0.7431	-0.0139
Client 7	0.7039	0.7829	+0.0789
Client 8	0.7292	0.7847	+0.0556

However, the improvements generated when personalizing models for each individual on Dataset 2b will be less than those generated for Dataset 2a because Dataset 2b's binary classification problem is less complex, and thus, the global federated model's overall performance is relatively strong before the model is personalized. Yet, the overall performance of the global federated model has improved for many of the clients following personalizing their models (e.g., Clients 7 and 8), resulting in the overall accuracies for Dataset 2a increasing to **68.30%** and Dataset 2b increasing to **69.08%**.

4.5 PROPOSED FRAMEWORK: FEDADAPT WITH RIEMANNIAN MANIFOLD INTEGRATION RESULT ANALYSIS

The Federated Learning Model for EEG will be improved by integrating Riemannian manifold-based feature alignment and aggregation with the FedAdapt training strategy to better capture the intrinsic geometric structure of covariance-based EEG features and to assist in converging to the same global model across heterogeneous clients. The framework proposed has two key components:

- 1) Riemannian Alignment (RA), which aligns covariance representations between clients prior to aggregation.
- 2) Riemannian Aggregation, which aggregates the model updates using a geometry-aware approach.

As with the traditional FedAdapt strategy, there are two phases for training:

- Phase 1: Federated Training for 30 Communication Rounds.
- Phase 2: Personalized Learning for 20 Epochs.

This design allows the model to learn global representations of EEG while adapting to each subject's unique neural patterns. During the Federated Training Phase, the global model improved with each complete Communication Round of Riemannian Alignment and Riemannian Aggregation, as shown in Table 4.5.

Table 4.5 - Federated Training Results with Riemannian Aggregation For both datasets

Round	BCI Competition 2a Dataset		BCI Competition 2b Dataset	
	Global Loss	Global Accuracy	Global Loss	Global Accuracy
1	0.9912	0.3209	0.5818	0.5844
5	0.6025	0.4598	0.5533	0.6465
10	0.5221	0.4397	0.5466	0.6679
15	0.4861	0.4655	0.5408	0.6718
20	0.4651	0.5096	0.5390	0.6595
25	0.4532	0.5402	0.5361	0.6603
30	0.4453	0.5584	0.5341	0.6365

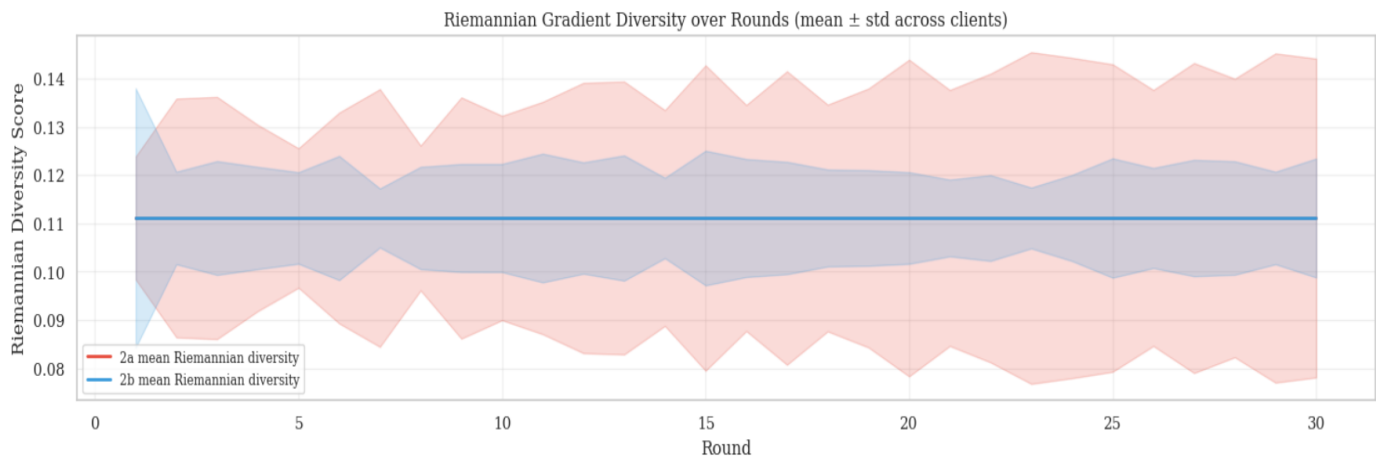


Figure 4.3 - Riemannian Gradient Density Over Rounds

In terms of convergence, the proposed method of using Riemannian-based aggregation outperformed classical federated aggregation methods significantly. For round 1, the global accuracy across all participants was 32.09% and for round 30 the accuracy improved to 55.84%. Riemannian Alignment allows for greater consistency across diverse EEG Client types when performing Federated Learning. The gradient density over rounds is shown in Figure 4.3. Client Personalization Results on Dataset 2a by Proposed framework before and after phase 2 is shown in Table 4.6 and Figure 4.5. For Dataset 2b from Table 4.7 and Figure 4.7, the global model developed by using the proposed methodology had an accuracy of 63.65% and exhibited stable convergence at 30 communication rounds. And the quality components of 2b is shown in Figure 4.6.

Table 4.6 - Client Personalization Results on Dataset 2a by Proposed framework
before and after PHASE 2

Clients	Accuracy before Personalization	Accuracy after Personalization	Improvement
Client 0	0.8103	0.8534	+0.0431
Client 1	0.4741	0.5431	+0.0690
Client 2	0.7155	0.8879	+0.1724
Client 3	0.4655	0.6552	+0.1897
Client 4	0.3707	0.5000	+0.1293
Client 5	0.3707	0.5172	+0.1466
Client 6	0.6121	0.8362	+0.2241
Client 7	0.8362	0.8190	-0.0172
Client 8	0.3707	0.8017	+0.4310

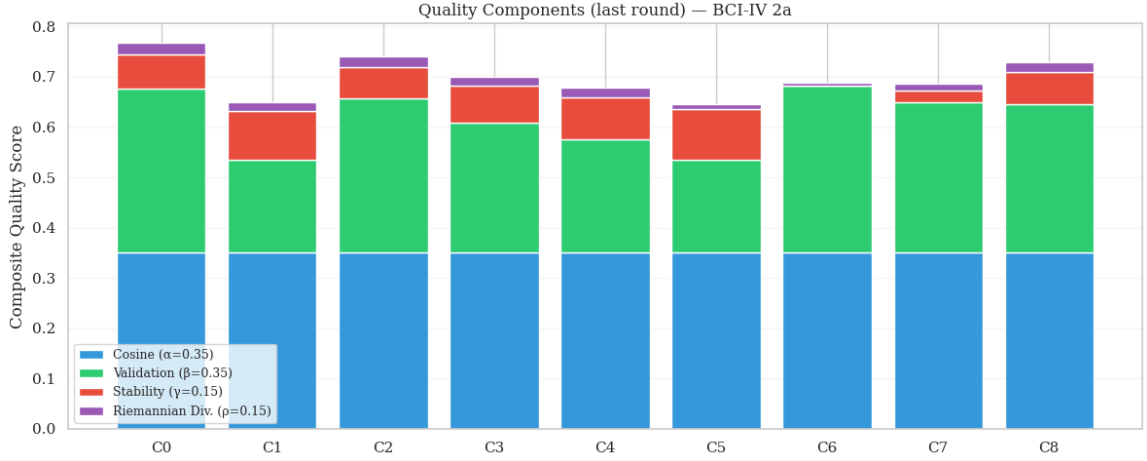


Figure 4.4 - Quality Components for BCI 2a Dataset

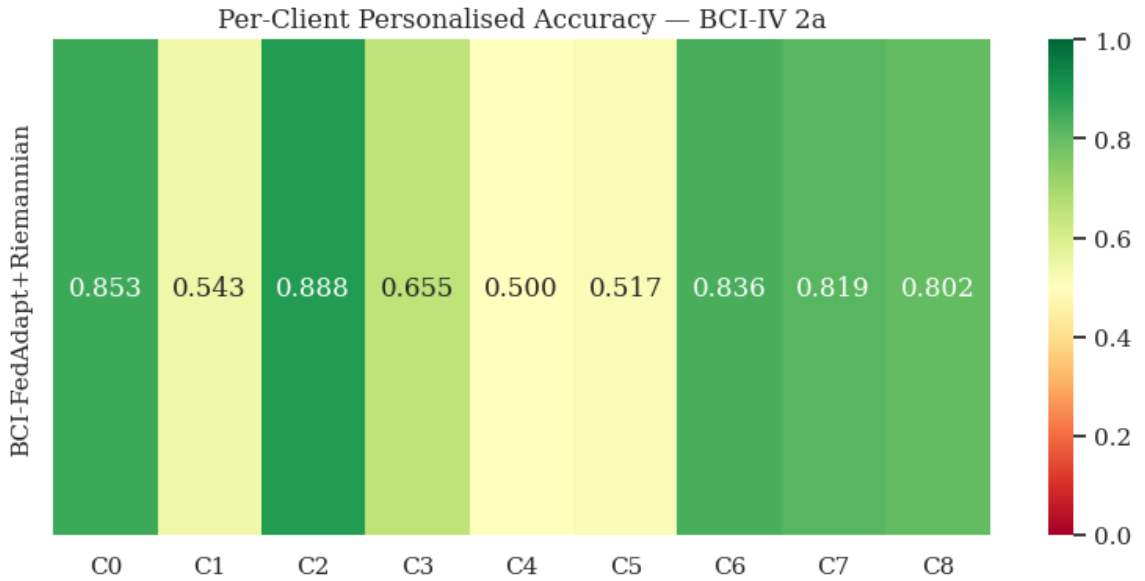


Figure 4.5 - Per client Personalized Accuracy - 2a Dataset

The Personalized phase of the study has further supported improvements in subject-specific performance; the overall accuracy for all Clients increased from **55.84%** to **71.26%** and the quality components are represented in Figure 4.4, which is a difference of 15.42%. Client 8's improvement was notable as their accuracy increased greater than any other Client; the accuracy of Client 8 increased by 43.10% illustrating the proposed methodology's adaptability to clients with diverse EEG Patterns.

Table 4.7 - Client Personalization Results on Dataset 2b by Proposed framework
before and after PHASE 2

Clients	Accuracy before Personalization	Accuracy after Personalization	Improvement
Client 0	0.5903	0.7778	+0.1875
Client 1	0.5956	0.5956	+0.0000
Client 2	0.5000	0.5000	+0.0000
Client 3	0.7027	0.8919	+0.1892
Client 4	0.5878	0.7635	+0.1757
Client 5	0.5417	0.6250	+0.0833
Client 6	0.7569	0.7708	+0.0139
Client 7	0.7434	0.7697	+0.0263
Client 8	0.7014	0.8333	+0.1319

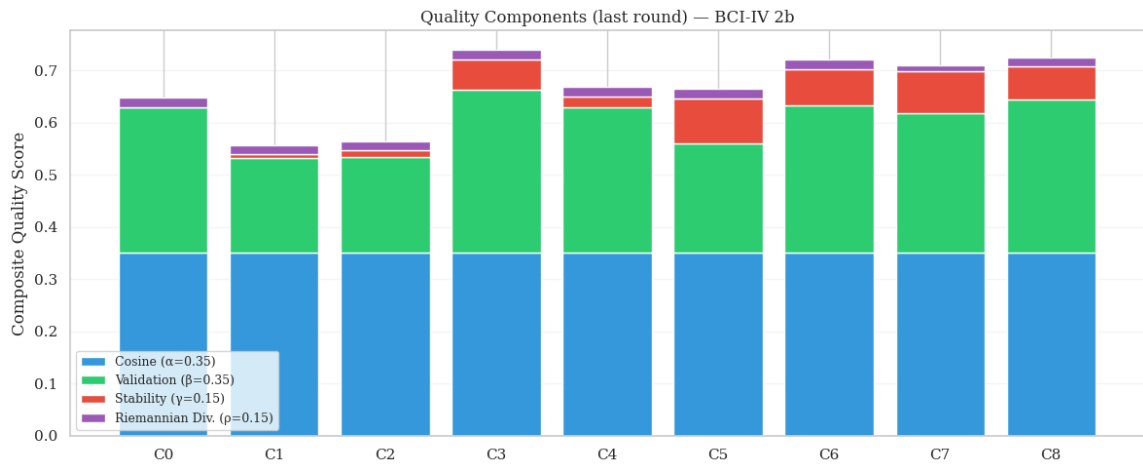


Figure 4.6 - Quality Components for BCI 2b Dataset

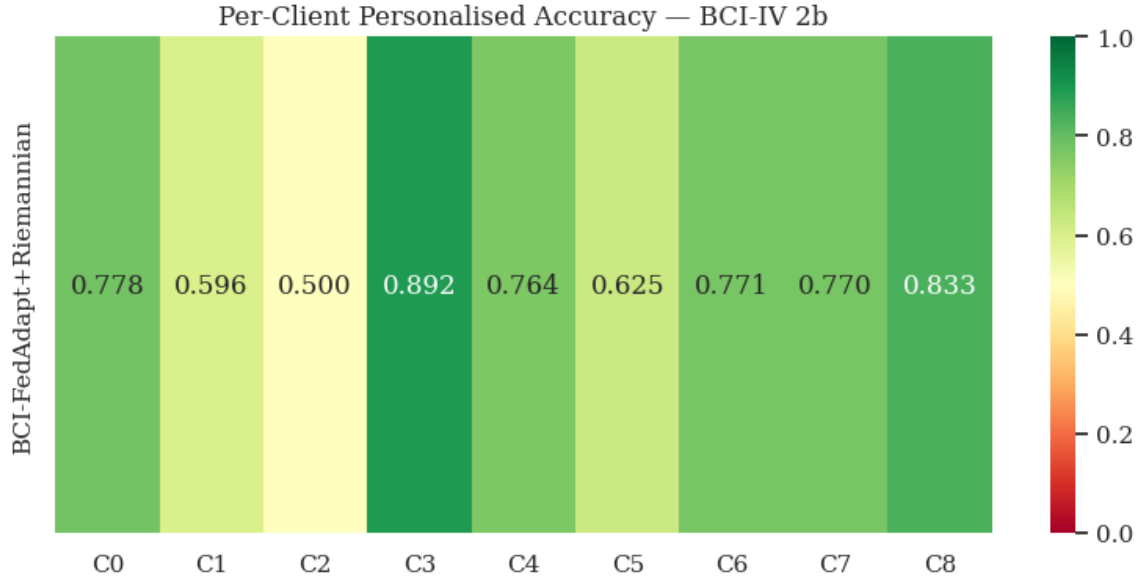


Figure 4.7 - Per client Personalized Accuracy - 2b Dataset

After personalization, the overall accuracy of the global aggregation method improved from 63.65% to 72.70%, a difference of 9.05%; the largest accuracy improvements were attributed to differences in Client 3 and Client 4 suggesting that the proposed methodology showed effective adaptability to the unique characteristics of each Client's EEG Signals.

3.6 DISCUSSION

The experiments showed that Riemannian manifold alignment and aggregation have been shown to enhance the performance of federated EEG models by using the FedAdapt framework for collaborative learning. The performance of the proposed framework on the more complicated Dataset 2a is much better than the alternatives, with global models converging much quicker and significantly greater personalization gains being achieved by clients compared to using other methods. Improvements in Dataset 2b are less pronounced but still achieved due to being much simpler in nature (a binary classification task). The development of Riemannian geometry in conjunction with federated deep learning has resulted in more effective means for treating heterogeneous EEG data while still ensuring subject privacy. The research findings presented in this study support the proposed integration model for decentralized Brain-Computer Interface Systems (BCI).

Chapter 5

Conclusion & Future works

We proposed a federated learning paradigm for analyzing EEG signals related to motor imagery tasks by combining Riemannian manifold-based feature extraction with an attention-based temporal convolutional network (ATCNet). The objectives of this study were to develop a solution to three critical problems faced in EEG-based brain-computer interface (BCI) systems: (i) variability in subjects' brain signals; (ii) concerns about privacy arising from the collection of centralized EEG data; and (iii) the distributed nature of data from different subjects' brains. To address these issues, EEG trials were first processed to produce covariance matrices representing the spatial relationships between EEG electrodes. Because covariance matrices are symmetric positive definite (SPD) matrices, Riemannian manifold-based methods were then used to transform the input covariance matrices from Riemannian manifold representations into tangent vector representations that can be used by machine learning algorithms. This geometrical feature extraction approach retains the inherent geometrical nature of EEG covariance data while producing improved feature representations than using traditional Euclidean geometric approaches.

After transforming the feature representations, we used them as input to the ATCNet which captures temporal dependencies and discriminative information from the EEG signals recorded during motor imagery tasks. The mechanism by which attention is employed allows ATCNet to focus on relevant temporal segments of the EEG signal, thereby enhancing the ability to classify motor imagery tasks. The training of the model utilized federated learning strategies to support collaborative learning without violating data privacy with the model trained on a federated learning basis (individual subjects were treated as separate clients). Rather than sharing raw EEG data between clients, clients trained and developed model updates locally and then sent those updates to the server. The method of aggregation was named FedAdapt, where aggregation was done in a manner that supported efficient global learning and also allowed certain layer(s) of the model to adapt

based on subject-specific characteristics as part of the personalization phase. The use of FedAdapt, therefore, reduced the negative effect on model performance caused by subjects having non-identical distributions of EEG data. Results showed that the framework successfully learned valuable, global, representative features and also enhanced subject-specific performance by providing personalized updates to the global model. An ablation study on the various aggregation methods indicated the findings of this study show the proposed aggregation method to be much better suited when dealing with heterogeneous datasets such as EEG.

The overall results of this study have shown that the combination of Riemannian manifold-based feature extraction, using deep neural networks (especially for classifying motor imagery of many subjects simultaneously) in addition to using federated learning creates a very robust approach to motor imagery classification while also creating a means to maintain data privacy.

While the suggested framework represents an encouraging direction for future research, other areas of exploration continue to exist. For example: The creation of enhanced personalized federated learning approaches that can accommodate global knowledge sharing along with subject-specific adaptations would provide additional opportunities for enhancing the overall quality of the data collected through these methodologies. Adaptive client weighting or dynamic model partitioning techniques could help achieve more rapid or improved global convergence rates in highly heterogeneous EEG settings. Additionally, generating improved efficiencies within both computational power and communication methodologies would result in greater scalability of federated EEG learning systems with respect to their applicability for real-world deployment within decentralized BCI systems.

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Appendices

<

APPENDIX A – SOURCE CODE IMPLEMENTATION

```
import warnings, copy, random, os
warnings.filterwarnings('ignore')

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
import seaborn as sns
from typing import Dict, List, Tuple, Optional
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import DataLoader, TensorDataset, WeightedRandomSampler
from torch.optim.lr_scheduler import CosineAnnealingLR
from scipy.signal import butter, filtfilt, iirnotch
from scipy.linalg import logm, expm, sqrtm
import moabb

from moabb.datasets import BNCI2014_001, BNCI2014_004
from moabb.paradigms import MotorImagery
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

from pyriemann.estimation import Covariances
from pyriemann.utils.mean import mean_riemann
from pyriemann.utils.distance import distance_riemann
from pyriemann.utils.geodesic import geodesic_riemann

moabb.set_log_level('WARNING')

SEED = 42
torch.manual_seed(SEED); np.random.seed(SEED); random.seed(SEED)
DEVICE = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
os.makedirs('/mnt/user-data/outputs', exist_ok=True)
```

```
print(f'Device: {DEVICE}')
```

```
CFG_2A = dict(  
    dataset_name='BCI-IV 2a',  
    sfreq=250, tmin=0.0, tmax=4.0, fmin=4.0, fmax=40.0,  
    n_classes=4, n_channels=22,  
    notch_freq=50.0, apply_car=True,  
  
    apply_riemann_align=True,  
    apply_ea=False,  
    ea_eps=1e-6, norm_epochs=True,  
    window_size=500, step_size=125,  
    F1=16, D=2, F2=32, dropout=0.4,  
    n_heads=4, attn_dp=0.4, tcn_filters=32, tcn_kernels=4, tcn_dp=0.3,  
    local_epochs=10, batch_size=64, lr=3e-3, wd=1e-4,  
    grad_clip=1.0, label_smoothing=0.05, eta_min=1e-5,  
    aug_prob=0.5, aug_noise_std=0.01,  
    num_rounds=30, proximal_mu=0.02, scaffold_lr=3e-3, momentum_beta=0.9,  
  
    qa_alpha=0.35, qa_beta=0.35, qa_gamma=0.15, qa_riemann=0.15,  
    qa_temperature=0.5,  
    personal_epochs=20, personal_lr=5e-4, personal_wd=1e-5,  
  
    use_riemann_agg=True,  
    riemann_agg_layers=['fc'],  
)  
print('CFG_2A ready')
```

```
CFG_2B = dict(  
    dataset_name='BCI-IV 2b',  
    sfreq=250, tmin=0.0, tmax=4.0, fmin=8.0, fmax=30.0,  
    n_classes=2, n_channels=3,  
    apply_car=True,  
    apply_riemann_align=True,  
    apply_ea=False,
```

```

    ea_eps=1e-6, norm_epochs=True,
    window_size=500, step_size=125,
    F1=8, D=2, F2=16, dropout=0.35,
    n_heads=2, attn_dp=0.3, tcn_filters=16, tcn_kernels=4, tcn_dp=0.2,
    local_epochs=10, batch_size=64, lr=3e-3, wd=1e-4,
    grad_clip=1.0, label_smoothing=0.1, eta_min=1e-5,
    aug_prob=0.5, aug_noise_std=0.01,
    num_rounds=30, proximal_mu=0.02, scaffold_lr=3e-3, momentum_beta=0.9,
    qa_alpha=0.35, qa_beta=0.35, qa_gamma=0.15, qa_riemann=0.15,
    qa_temperature=0.5,
    personal_epochs=20, personal_lr=5e-4, personal_wd=1e-5,
    use_riemann_agg=True,
    riemann_agg_layers=['fc'],
)
print('CFG_2B ready')

```

```

def euclidean_alignment(X: np.ndarray, eps: float = 1e-6) -> np.ndarray:

```

```

    N, C, T = X.shape
    R = np.zeros((C, C), dtype=np.float64)
    for i in range(N):
        xi = X[i].astype(np.float64)
        R += xi @ xi.T / T
    R /= N
    R += eps * np.eye(C)
    eigvals, eigvecs = np.linalg.eigh(R)
    eigvals = np.maximum(eigvals, eps)
    R_inv_sqrt = eigvecs @ np.diag(1.0 / np.sqrt(eigvals)) @ eigvecs.T
    return np.einsum('ij,njt->nit', R_inv_sqrt, X).astype(np.float32)

```

```

def normalize_epochs(X: np.ndarray) -> np.ndarray:

```

```

    mu = X.mean(axis=2, keepdims=True)
    std = X.std(axis=2, keepdims=True) + 1e-8
    return (X - mu) / std

```



```

def preprocess(X: np.ndarray, cfg: dict) -> np.ndarray:

    if cfg.get('apply_car', True):
        X = apply_car(X)
    if cfg.get('apply_riemann_align', False):
        X = riemannian_alignment(X, eps=cfg.get('ea_eps', 1e-6))
    elif cfg.get('apply_ea', True):
        X = euclidean_alignment(X, eps=cfg.get('ea_eps', 1e-6))
    if cfg.get('norm_epochs', True):
        X = normalize_epochs(X)
    return X

def load_and_preprocess(cfg: dict) -> Tuple[dict, int, object]:
    name = cfg.get('dataset_name', 'BCI-IV 2a')
    align_tag = 'Riemannian' if cfg.get('apply_riemann_align') else 'Euclidean'
    print(f>Loading {name} [{align_tag} alignment] ...')
    if '2a' in name or '001' in name:
        dataset = BNCI2014_001()
    else:
        dataset = BNCI2014_004()
    paradigm = MotorImagery(
        n_classes=cfg['n_classes'], fmin=cfg['fmin'], fmax=cfg['fmax'],
        tmin=cfg['tmin'], tmax=cfg['tmax'], resample=cfg['sfreq'])
    X, y, meta = paradigm.get_data(dataset=dataset)
    le = LabelEncoder()
    y_enc = le.fit_transform(y)
    print(f' Raw shape: {X.shape} Classes: {list(le.classes_)}')
    cdata = {}
    for cid, subj in enumerate(sorted(meta['subject'].unique())):
        mask = (meta['subject'] == subj).values
        Xs, ys = preprocess(X[mask].astype(np.float32), cfg), y_enc[mask].astype(np.int64)
        Xtr, Xte, ytr, yte = train_test_split(
            Xs, ys, test_size=0.2, stratify=ys, random_state=SEED)
        cdata[cid] = dict(X_train=Xtr, X_test=Xte, y_train=ytr, y_test=yte)
        print(f' Client {cid:02d}: train={len(ytr)} test={len(yte)}')
    n_times = X.shape[-1]
    return cdata, n_times, le.classes_

```

```
print('load_and_preprocess defined')
```

```
class TCNBlock(nn.Module):
```

```
    def __init__(self, in_ch, out_ch, kernel_size=4, dilation=1, dp=0.3):
        super().__init__()
        self.ks, self.dil = kernel_size, dilation
        self.conv1 = nn.Conv1d(in_ch, out_ch, kernel_size, dilation=dilation)
        self.conv2 = nn.Conv1d(out_ch, out_ch, kernel_size, dilation=dilation)
        self.bn1 = nn.BatchNorm1d(out_ch); self.bn2 = nn.BatchNorm1d(out_ch)
        self.act = nn.ELU(); self.drop = nn.Dropout(dp)
        self.residual = nn.Conv1d(in_ch, out_ch, 1) if in_ch != out_ch else nn.Identity()
```

```
    def _pad(self, x):
```

```
        total = self.dil * (self.ks - 1)
        return F.pad(x, (total // 2, total - total // 2))
```

```
    def forward(self, x):
```

```
        T = x.size(2); res = self.residual(x)
        h = self.drop(self.act(self.bn1(self.conv1(self._pad(x))[:, :, :T])))
        h = self.drop(self.act(self.bn2(self.conv2(self._pad(h))[:, :, :T])))
        return h + res
```

```
class ATCNet(nn.Module):
```

```
    def __init__(self, n_classes, n_channels, n_times, F1=16, D=2, F2=32, dp=0.4,
                    n_heads=4, attn_dp=0.4, tcn_filters=32, tcn_kernels=4, tcn_dp=0.3):
        super().__init__()
        self.temporal = nn.Sequential(
            nn.Conv2d(1, F1, (1, 64), padding=(0, 32), bias=False), nn.BatchNorm2d(F1))
        self.spatial = nn.Sequential(
            nn.Conv2d(F1, D*F1, (n_channels, 1), groups=F1, bias=False),
            nn.BatchNorm2d(D*F1), nn.ELU(), nn.AvgPool2d((1, 4)), nn.Dropout(dp))
        self.separable = nn.Sequential(
            nn.Conv2d(D*F1, D*F1, (1, 16), padding=(0, 8), groups=D*F1, bias=False),
            nn.Conv2d(D*F1, F2, (1, 1), bias=False),
```

```

        nn.BatchNorm2d(F2), nn.ELU(), nn.AvgPool2d((1, 8)), nn.Dropout(dp))
with torch.no_grad():
    _x = torch.zeros(1, 1, n_channels, n_times)
    self.T_feat = self.separable(self.spatial(self.temporal(_x))).shape[-1]
    self.attn_norm = nn.LayerNorm(F2)
    self.mha = nn.MultiheadAttention(F2, n_heads, dropout=attn_dp, batch_first=True)
    self.attn_drop = nn.Dropout(attn_dp)
    self.tcn1 = TCNBlock(F2, tcn_filters, tcn_kernels, dilation=1, dp=tcn_dp)
    self.tcn2 = TCNBlock(tcn_filters, tcn_filters, tcn_kernels, dilation=2, dp=tcn_dp)
    self.out_norm = nn.LayerNorm(tcn_filters)
    self.fc = nn.Linear(tcn_filters, n_classes)

def forward(self, x):
    x = x.unsqueeze(1)
    x = self.separable(self.spatial(self.temporal(x)))
    x = x.squeeze(2).transpose(1, 2)
    res = x
    x, _ = self.mha(self.attn_norm(x), self.attn_norm(x), self.attn_norm(x))
    x = self.attn_drop(x) + res
    x = x.transpose(1, 2)
    x = self.tcn2(self.tcn1(x))
    return self.fc(self.out_norm(x.mean(dim=2)))

class BCIFedAdaptClient:
    def __init__(self, client_id, data, cfg):
        self.cid = client_id; self.cfg = cfg
        self.n_train = len(data['y_train'])
        ws, ss = cfg['window_size'], cfg['step_size']
        X_win, y_win = make_windows(data['X_train'], data['y_train'], ws, ss)
        Xt, yt = torch.from_numpy(X_win), torch.from_numpy(y_win)
        counts = np.bincount(y_win)
        weights = 1.0 / counts[y_win]
        sampler = WeightedRandomSampler(torch.from_numpy(weights).float(), len(weights))
        self.balanced_loader = DataLoader(TensorDataset(Xt, yt),
                                          batch_size=cfg['batch_size'],
                                          sampler=sampler, drop_last=True)
        self.X_test_raw = data['X_test']; self.y_test = data['y_test']

```

```

self.model = make_model(cfg)
self.criterion = nn.CrossEntropyLoss(label_smoothing=cfg.get('label_smoothing', 0.0))
self._grad_norms: List[float] = []

def get_universal_state(self):
    return {k: v.clone() for k, v in self.model.state_dict().items() if is_universal(k)}

def set_universal_state(self, state):
    current = self.model.state_dict()
    current.update({k: v.to(DEVICE) for k, v in state.items()})
    self.model.load_state_dict(current)

def evaluate(self):
    ws, ss = self.cfg['window_size'], self.cfg['step_size']
    preds = soft_vote_predict(self.model, self.X_test_raw, ws, ss)
    return accuracy_score(self.y_test, preds), len(self.y_test)

def local_train(self, strategy='BCI-FedAdapt'):
    self._grad_norms = []
    self.model.train()
    opt = torch.optim.Adam(self.model.parameters(),
                           lr=self.cfg['lr'], weight_decay=self.cfg['wd'])
    sched = CosineAnnealingLR(opt, T_max=self.cfg['local_epochs'],
                              eta_min=self.cfg['eta_min'])
    total_loss, steps = 0.0, 0
    for _ in range(self.cfg['local_epochs']):
        for Xb, yb in self.balanced_loader:
            Xb, yb = Xb.to(DEVICE), yb.to(DEVICE)
            if random.random() < self.cfg.get('aug_prob', 0):
                Xb = Xb + torch.randn_like(Xb) * self.cfg.get('aug_noise_std', 0.01)
            opt.zero_grad()
            loss = self.criterion(self.model(Xb), yb)
            loss.backward()
            gn = sum(p.grad.data.norm(2).item() ** 2 for p in self.model.parameters()
                    if p.grad is not None) ** 0.5
            self._grad_norms.append(gn)
            nn.utils.clip_grad_norm_(self.model.parameters(), self.cfg['grad_clip'])

```

```

        opt.step()
        total_loss += loss.item(); steps += 1
    sched.step()
    return total_loss / max(steps, 1)

def personalise(self):
    for name, param in self.model.named_parameters():
        param.requires_grad = is_personal(name)
    personal_params = [p for p in self.model.parameters() if p.requires_grad]
    if not personal_params:
        return self.evaluate()[0]
    opt = torch.optim.Adam(personal_params,
                            lr=self.cfg['personal_lr'], weight_decay=self.cfg['personal_wd'])
    sched = CosineAnnealingLR(opt, T_max=self.cfg['personal_epochs'],
                              eta_min=self.cfg['eta_min'])
    self.model.train()
    for _ in range(self.cfg['personal_epochs']):
        for Xb, yb in self.balanced_loader:
            Xb, yb = Xb.to(DEVICE), yb.to(DEVICE)
            opt.zero_grad()
            loss = self.criterion(self.model(Xb), yb)
            loss.backward()
            nn.utils.clip_grad_norm_(personal_params, self.cfg['grad_clip'])
            opt.step()
        sched.step()
    for p in self.model.parameters():
        p.requires_grad = True
    return self.evaluate()[0]

```

```

def aggregate_bci_fedadapt(clients, w_univ_snapshot, cfg, round_num):

```

```

    weights, diag = compute_quality_weights(clients, w_univ_snapshot, cfg)
    use_riemann = cfg.get('use_riemann_agg', False)
    riemann_layers = cfg.get('riemann_agg_layers', [])

    agg_state = {}

```

```

for key in clients[0].get_universal_state().keys():
    tensors = [c.get_universal_state()[key].float() for c in clients]

    apply_riem = use_riemann and any(key.startswith(pfx) for pfx in riemann_layers)

    if apply_riem and tensors[0].dim() == 2:

        mats = []
        for t in tensors:
            v = t.cpu().numpy().reshape(-1)
            S = np.outer(v, v)
            norm = np.linalg.norm(S, 'fro') + 1e-12
            S = S / norm + 1e-5 * np.eye(len(v))

            if len(v) > 64:
                np.random.seed(round_num)
                P = np.random.randn(len(v), 64)
                P /= np.linalg.norm(P, axis=0, keepdims=True)
                v_p = P.T @ v
                S = np.outer(v_p, v_p) / (np.linalg.norm(v_p)**2 + 1e-12)
                S += 1e-4 * np.eye(64)
            mats.append(S)

        try:
            M_bar = mean_riemann(np.array(mats), tol=1e-6, maxiter=50)

            eigvals, eigvecs = np.linalg.eigh(M_bar)
            principal = eigvecs[:, -1]

            flat_avg = sum(w * t.cpu().numpy() for w, t in zip(weights, tensors))
            flat_norm = np.linalg.norm(flat_avg)
            if flat_avg.ndim == 2 and principal.shape[0] in flat_avg.shape:

                d_out, d_in = flat_avg.shape
                if principal.shape[0] == d_out:
                    reconstructed = np.outer(principal, np.ones(d_in)) * flat_norm / (d_in ** 0.5)
                    alpha_blend = 0.3

```

```

        blended = (1 - alpha_blend) * flat_avg + alpha_blend * reconstructed
    else:
        blended = flat_avg
    agg_state[key] = torch.tensor(blended, dtype=torch.float32).to(DEVICE)
else:
    agg_state[key] = torch.tensor(flat_avg, dtype=torch.float32).to(DEVICE)
except Exception:

    flat_avg = sum(w * t for w, t in zip(weights, tensors))
    agg_state[key] = flat_avg.to(DEVICE)
else:

    agg_state[key] = sum(w * t for w, t in zip(weights, tensors)).to(DEVICE)

for client in clients:
    client.set_universal_state(agg_state)
return agg_state, weights, diag

def plot_quality_components(results, cfg, tag, save):

    key = 'BCI-FedAdapt+Riemannian'
    if key not in results: key = list(results.keys())[0]
    last_diag = results[key]['diag_log'][-1]
    n = len(last_diag['cos'])
    x = np.arange(n)
    a, b, g, r = cfg['qa_alpha'], cfg['qa_beta'], cfg['qa_gamma'], cfg.get('qa_riemann', 0)
    fig, ax = plt.subplots(figsize=(12, 5))
    ax.bar(x, a * last_diag['cos'], label=f'Cosine ( $\alpha=\{a\}$ ), color='#3498db')
    ax.bar(x, b * last_diag['val'], bottom=a * last_diag['cos'],
           label=f'Validation ( $\beta=\{b\}$ ), color='#2ecc71')
    base2 = a * last_diag['cos'] + b * last_diag['val']
    ax.bar(x, g * last_diag['stab'], bottom=base2,
           label=f'Stability ( $\gamma=\{g\}$ ), color='#e74c3c')
    base3 = base2 + g * last_diag['stab']
    ax.bar(x, r * last_diag['riemann'], bottom=base3,
           label=f'Riemannian Div. ( $\rho=\{r\}$ ), color='#9b59b6')
    ax.set_xticks(x); ax.set_xticklabels([f'C {i}' for i in range(n)])

```

```

ax.set_ylabel('Composite Quality Score'); ax.set_title(f'Quality Components (last round) — {tag}')
ax.legend(fontsize=9); ax.grid(alpha=0.2, axis='y')
plt.tight_layout(); plt.savefig(save, dpi=150); plt.show()

```

```

def plot_personalisation_gain(results, cfg, tag, save):
    key = 'BCI-FedAdapt+Riemannian'
    if key not in results: key = list(results.keys())[0]
    r = results[key]
    before = r['personal_accs_before']; after = r['personal_accs_after']
    deltas = [a - b for a, b in zip(after, before)]
    n = len(before); pal = sns.color_palette('tab10', n)
    x = np.arange(n)
    fig, axes = plt.subplots(1, 2, figsize=(14, 5))
    axes[0].bar(x-0.2, before, 0.38, label='Phase 1', color=[p+(0.6,) for p in pal], edgecolor='w')
    axes[0].bar(x+0.2, after, 0.38, label='Phase 2', color=pal, edgecolor='w')
    axes[0].set_xticks(x); axes[0].set_xticklabels([f'C {i}' for i in range(n)])
    axes[0].set_ylim(0, 1); axes[0].set_ylabel('Accuracy')
    axes[0].set_title(f'Phase 1 vs Phase 2 per Client — {tag}')
    axes[0].legend(); axes[0].grid(alpha=0.2, axis='y')
    axes[0].axhline(1/cfg['n_classes'], ls='--', color='grey', lw=1)
    cols = ['#27ae60' if d >= 0 else '#e74c3c' for d in deltas]
    axes[1].bar(x, deltas, color=cols, edgecolor='w')
    axes[1].axhline(0, color='black', lw=0.8)
    axes[1].set_xticks(x); axes[1].set_xticklabels([f'C {i}' for i in range(n)])
    axes[1].set_ylabel('Δ Accuracy (Phase2 – Phase1)')
    axes[1].set_title(f'Personalisation Gain — {tag}')
    axes[1].grid(alpha=0.2, axis='y')
    plt.tight_layout(); plt.savefig(save, dpi=150); plt.show()

```

```

def plot_eval(final, strategies, cfg, class_names, tag, base_path):
    strats = strategies
    n_cls = cfg['n_classes']
    fig = plt.figure(figsize=(6*len(strats), 10))
    gs = gridspec.GridSpec(2, len(strats))
    accs = [final[s][0] for s in strats]

```



```

for col, s in enumerate(strats):
    ax_cm = fig.add_subplot(gs[0, col])
    _, yl, yp = final[s]
    cm = confusion_matrix(yl, yp)
    cm_norm = cm.astype(float) / cm.sum(axis=1, keepdims=True)
    sns.heatmap(cm_norm, annot=True, fmt='.2f', cmap='Blues',
                xticklabels=class_names, yticklabels=class_names,
                vmin=0, vmax=1, ax=ax_cm)

    ax_cm.set_title(f'{s}\nAcc={final[s][0]:.3f}', fontsize=9)
    ax_cm.set_xlabel('Predicted'); ax_cm.set_ylabel('True')
ax_bar = fig.add_subplot(gs[1, :])
pal = sns.color_palette('tab10', len(strats))
bars = ax_bar.bar(strats, accs, color=pal, edgecolor='white')
ax_bar.axhline(1.0/n_cls, ls='--', color='grey', lw=1, label=f'Chance ({1/n_cls:.2f})')
ax_bar.set_ylim(0, 1.1); ax_bar.set_ylabel('Global Test Accuracy')
ax_bar.set_title(f'Final Accuracy — {tag}')
for bar, a in zip(bars, accs):
    ax_bar.text(bar.get_x()+bar.get_width()/2, a+0.01, f'{a:.3f}',
                ha='center', va='bottom', fontsize=8)
ax_bar.legend(fontsize=9)
plt.tight_layout()
plt.savefig(f'{base_path}_eval.png', dpi=150)
plt.show()

final_2b = {}
for strat in STRATEGIES:
    ps = results_2b[strat]['final_personal_states']
    acc, yl, yp = eval_global(None, CFG_2B, client_data_2b, personal_states=ps)
    final_2b[strat] = (acc, yl, yp)
    print(f'{strat}: {acc:.4f}')
best_2b = max(final_2b, key=lambda s: final_2b[s][0])

plot_eval(final_2b, STRATEGIES, CFG_2B, CLASS_NAMES_2B,
          '/mnt/user-data/outputs/bci_iv_2b')
plot_per_client_heatmap(results_2b, STRATEGIES, CFG_2B, client_data_2b,
                        '/mnt/user-data/outputs/per_client_2b.png')
print_summary(final_2b, STRATEGIES, CFG_2B, results_2b, 'BCI-IV 2b', best_2b)

```

```

_, yl, yp = final_2b[best_2b]
print(classification_report(yl, yp, target_names=CLASS_NAMES_2B))

fig = plt.figure(figsize=(18, 10))
gs = gridspec.GridSpec(2, 3, figure=fig, hspace=0.45, wspace=0.35)

ax0 = fig.add_subplot(gs[0, 0])
x = np.arange(len(STRATEGIES))
w = 0.35
accs_2a = [final_2a[s][0] for s in STRATEGIES]
accs_2b = [final_2b[s][0] for s in STRATEGIES]
pal = sns.color_palette('tab10', len(STRATEGIES))
ax0.bar(x-w/2, accs_2a, w, label='BCI-IV 2a (4-class)',
        color=[p+(0.8,) for p in pal], edgecolor='white')
ax0.bar(x+w/2, accs_2b, w, label='BCI-IV 2b (2-class)',
        color=pal, edgecolor='white')
ax0.axhline(0.25, ls=':', color='#e74c3c', lw=1.2, label='2a chance')
ax0.axhline(0.50, ls=':', color='#3498db', lw=1.2, label='2b chance')
ax0.set_xticks(x)
ax0.set_xticklabels([s.replace('BCI-FedAdapt+', 'FA+\n') for s in STRATEGIES],
                    rotation=20, ha='right', fontsize=8)
ax0.set_ylabel('Global Test Accuracy'); ax0.set_ylim(0, 1.1)
ax0.set_title('Final Accuracy — 2a vs 2b'); ax0.legend(fontsize=7)
for bar, a in zip(ax0.patches[:len(STRATEGIES)], accs_2a):
    ax0.text(bar.get_x()+bar.get_width()/2, a+0.01, f'{a:.3f}',
            ha='center', va='bottom', fontsize=7)

print('\n' + '='*72)
print(' FINAL RESULTS SUMMARY — BCI-FedAdapt + Riemannian Manifold')
print('='*72)
for ds_tag, final_, cfg_, results_ in [
    ('BCI-IV 2a (4-class)', final_2a, CFG_2A, results_2a),
    ('BCI-IV 2b (2-class)', final_2b, CFG_2B, results_2b)]:
    chance = 1.0 / cfg_['n_classes']

```

```

print(f'\n {ds_tag}')
print(f' {"—"*60}')
for s in STRATEGIES:
    acc = final_[s][0]
    p1 = results_[s]['mean_acc_phase1']
    p2 = results_[s]['mean_acc_phase2']
    riemann_q = np.mean([d['riemann'].mean() for d in results_[s]['diag_log']])
    print(f' Strategy : {s}')
    print(f' Phase 1 : {p1:.4f} Phase 2: {p2:.4f} Global: {acc:.4f}')
    print(f' Δ chance : {acc - chance:+.4f} Mean Riemannian diversity: {riemann_q:.5f}')

```

APPENDIX B: HYPERPARAMETER TUNING

The Propose FedAdapt framework's ATCNet architecture was developed to allow stable training and good performance by tuning hyperparameters. Many training parameters which impact the training process changed as per experiments, including the learning rate, the communication rounds, the number of local training epochs, and the batch size. The learning rate is used to control the rate at which the parameters are updated during the training. Various learning rates were tried and it was found that a learning rate of 0.001 provided stable convergence for all experiments. The number of federated communication rounds is indicates how many times the global model is updated; specifically, it is how many times the local models are aggregated to form the global model. In this research study, 30 communication rounds were completed to have enough global information to learn the global EEG pattern. Each client has 10 local training epochs for every federated training phase to keep balance between local learning and global model synchronization before sending updates back to the server. There were 20 local training epochs completed for the personalization phase of the model to help the model learn the subject specific characteristics of the EEG data. A batch size was established at 32 in order to have good use of the GPU without impacting the quality of training. The Adam optimizer was selected as the optimizer for the model due to its adaptive nature and its performance on deep learning applications.

APPENDIX C: OTHER REFERENCES

<https://bnci-horizon-2020.eu/database/data-sets>

<https://www.sciencedirect.com/topics/medicine-and-dentistry/motor-imagery>

<https://flower.ai/docs/framework/tutorial-series-what-is-federated-learning.html>